

Legends of SCIENCE

Book 3

Anne Clark



Legends of

SCIENCE

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Anne Clark

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ABN 50 074 266 023

108 Main Road

Clayton South VIC 3169

Ph: +61 3 9558 4433

info@blake.com.au

www.blake.com.au

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Introduction

People have questioned the world around them for thousands of years, leading to an accumulation of scientific knowledge through the efforts of many. As Isaac Newton said, 'If I have seen further, it is by standing on the shoulders of giants.'

This series uses the work of individual scientists to give students a context in which to explore ideas from many branches of science, including biology, chemistry, geology and physics. Each worksheet explores the life and work of an eminent scientist, including those living and dead, female and male.

Teachers will find these worksheets useful for extension, revision, homework and assessment.

They will also engage student interest in the field of science, providing a solid base for students to explore further both the contributions of individual scientists and the concepts they investigated. Some activities require access to research materials to complete the tasks.

The text of this book is also on the CD-ROM attached to the inside back cover.

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Robert Goddard

1882–1945

- Robert Goddard was born in Massachusetts, USA. He received his Bachelor of Science degree from Worcester Polytechnic Institute and his Masters and PhD from Clark University.
- In 1912, he developed the mathematical theory of rocket propulsion, and proved that space flight was possible.
- In 1914, he patented the first multistage rocket, and launched the first liquid-fuelled rocket in 1926.
- From 1930–35, he launched rockets that attained speeds of up to 1425 kilometres per hour.
- During World War 1 and 2, Robert developed rocket-assisted take-off for aircraft carrier planes, rocket motors and the bazooka.
- The Goddard Space Flight Centre in Maryland, USA, was named in his honour.

History of rockets

- 1 Complete the information using these words: Cold War, Chinese, oxygen, gunpowder, rocket, London, range, missiles, man, space, astronaut.

The _____ invented _____ in the first century AD and used it for warfare.

In 1903, Konstantin Tsiolkovsky proposed the use of _____ and hydrogen as a propellant for rocket propulsion.

Robert Goddard developed the first liquid-fuelled _____ in 1926.

During World War 2, Wernher von Braun developed the V2 guided missile rocket which was used to bomb _____. It had a 500 kg warhead and an operational _____ of 300 km. After the war, the Americans worked with German scientists to build the first _____ rocket.

At the same time, the Soviet Union developed the R-7, which launched the first _____, satellite and planetary probe into space.

During the _____, intercontinental nuclear ballistic _____ were developed. By the end of the 1960s, Saturn V had landed the first _____ on the moon.

Rockets have had a huge impact on our lives through research, space travel, satellite telecommunications and their use as weapons of warfare.

- 2 Describe the benefits derived from rockets.
- 3 Why did the Soviet Union and America both want to land the first person on the moon?
- 4 Explain how guided missiles and nuclear weapons have contributed to global political tension.

Definitions

Rocket – Projectile that can travel through space

Multistage rocket – Rocket with several fuel sections that are released after use

V2 – Guided missile developed by the Germans to bomb London in World War 2

Propellant – Fuel used in a rocket

Saturn V – America’s moon rocket

Saturn V was taller than a 36 storey building and was the largest and most powerful rocket ever launched.

Wernher von Braun developed 32 of these rockets, and even though they contained more than one million parts, not a single one failed.

In July 1969, the Saturn V and the crew of Apollo 11 landed on the moon.

- 5 Refer to table 1 to answer these questions.
- a What amount of force was needed to push the rocket into space?
- b How much weight was lost in fuel and debris during initial take-off to the moon?
- c Explain why fuel is one of the largest weights in a rocket.

Table 1 – Features of Saturn V

Length	111 m
Payload to orbit	129 300 kg
Payload to moon	48 500 kg
Total thrust force	33 360 000 Newtons
Propellant	Liquid hydrogen and oxygen

Source: www.nasa.gov/home/index.html

The space shuttle

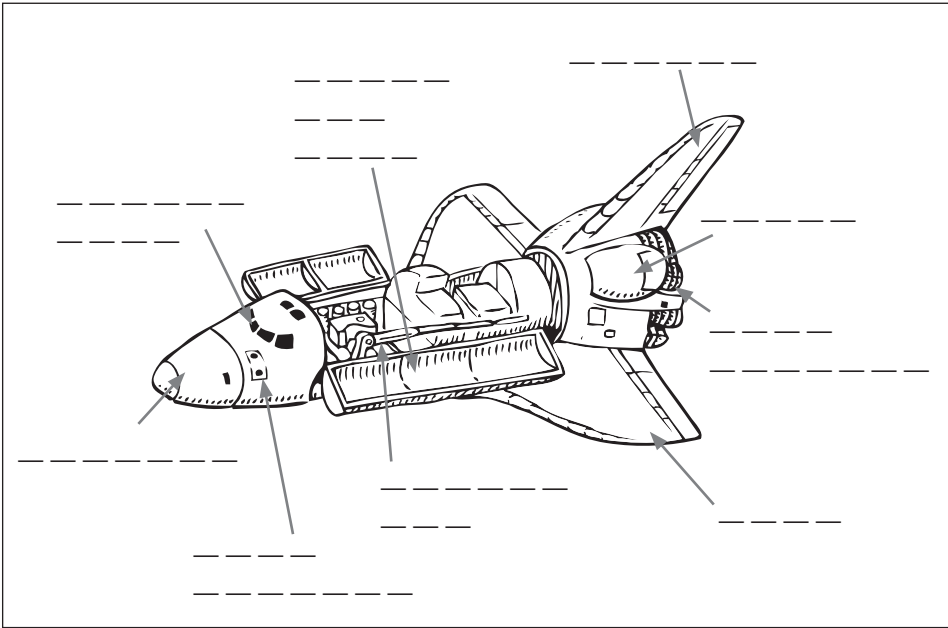
The space shuttle was first launched in 1981. It had the features of a rocket, aircraft and glider. It was a more efficient method of space travel because it could be reused, it could operate on land and in space, and be used for scientific, medical, commercial and military operations.

- 6 Label the figure 1 using these words: *engines, star tracker, flight deck, remote arm, cargo bay door, wing, tanks, main engines, rudder.*

Characteristics of the shuttle

- 7 Refer to table 2 to answer these questions.
- a At what speed can the space shuttle travel?
 - b What is its fuel source?
 - c What is the power of its main engine?
 - d How many people have been killed on space shuttle missions?

Figure 1 – Structure of the space shuttle



Satellites

The first satellite was called ‘Sputnik’ and was launched by the Soviet Union in 1957. There are now more than 2500 satellites that orbit Earth.

- 8 List examples of the uses of satellites in each of these areas.
- a Defence
 - b International voice and data communications
 - c Video conferencing
 - d Business
 - e Television
 - f Broadband Internet access
 - g Global positioning system
 - h Infrared astronomy
 - i Weather forecasting and monitoring
 - j Remote sensing

Table 2 – Characteristics of the shuttle

Structures	Orbiter, external tank, two solid rocket boosters and main engines
Number of astronauts	8
Time in space	Up to 2.5 weeks
Speed of more than	27 359 kilometres per hour
Main engine	Weights 1/7th as much as a train engine but delivers as much horsepower as 30 locomotives
Thrust	Three engines and two solid rocket boosters generate 3.3 million Newtons of thrust at lift-off.
Fuel	Liquid hydrogen stored at –253°C
Temperature in the engine combustion chamber	3316°C
Payload	24 990 kg
Accidents	Two – killed 14 crew

Source: www.nasa.gov/home/index.html

Fuel systems in rockets

Most rockets have an internal combustion engine and a solid or liquid propellant, or a combination of both. A chemical reaction between the fuel and the oxidiser causes an explosion that pushes the rocket upwards.

- 9 Draw a rocket fuel system. Use this word list to annotate the drawing to help explain the processes: *turbine, cooling pipe, hot exhaust gases, nozzle, waste gases, combustion chamber, turbo pump, liquid hydrogen, liquid oxygen.*

Extension

- * Describe the positive and negative impacts of rockets on global society.
- * What benefits can be gained from space research and travel?
- * Research Australia’s involvement in space exploration.
- * Find out what the Cold War was, how it started and how it ended.



Patricia Vickers-Rich

1944–

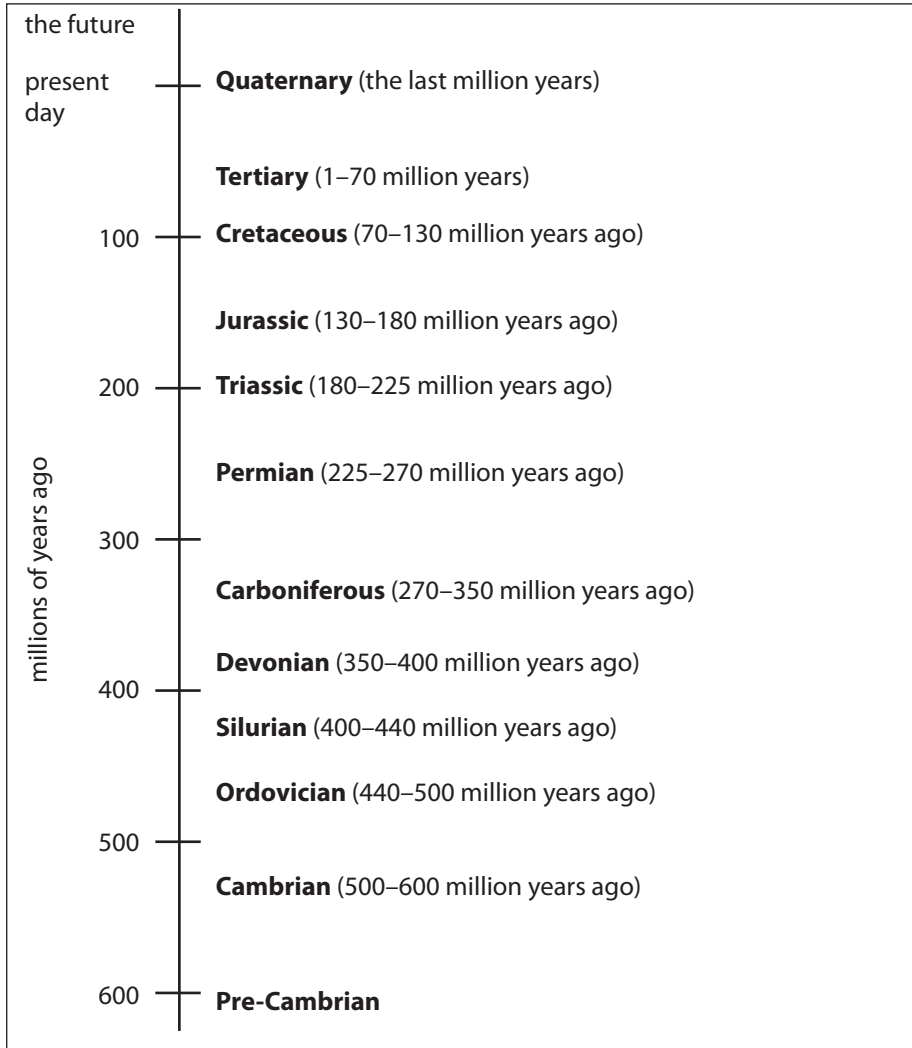
- Patricia was born in California, USA. In 1966, she obtained an Arts degree from the University of California and in 1973 a PhD from Columbia University.
- Patricia migrated to Australia in 1976 and became the founding director of the science centre at Monash University.
- Patricia researched the changes in Australia's plants and animals throughout the last 400 million years. She has written technical and popular books about dinosaurs.

Life on Earth

Life has existed on this planet for more than one billion years.

1 What were the earliest forms of life?

Figure 1 – Timeline of life appearing on Earth



2 Find out in which time periods the following organisms developed, and record this information on figure 1.

- | | | |
|---------|------------|-----------------|
| a fish | c mammals | e invertebrates |
| b birds | d bacteria | f reptiles |

3 How did the number and variety of species change over time?

Pangaea and continental drift

The Earth's crust is divided into plates which move over time. It is believed that all of the continents were once part of an ancient super continent called Pangaea.

4 Figure 2 shows the movement of the continents with time. Number the diagrams from 1 (earliest) to 5 (latest) to show the correct order.

Definitions

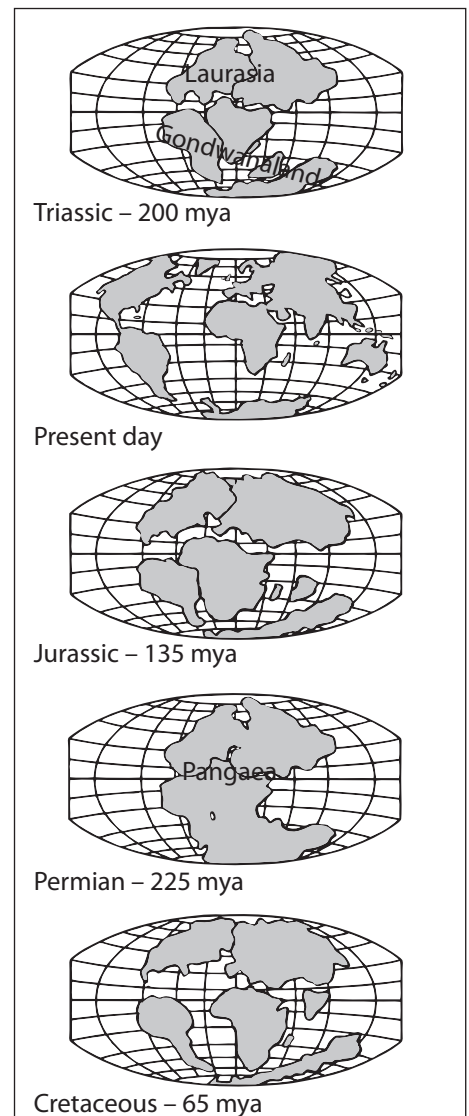
Fauna – Animals

Pangaea – Super continent believed to have existed 225 million years ago

Continental drift – Movement of continents over time

Evolution – Change in organisms over time due to natural selection

Figure 2 – Movement of the continents over time



- 5 Describe what happened to the land masses over time.
- 6 Which continents constituted Gondwana?
- 7 Which continent was closest to Australia?
- 8 How has the location of Australia changed over time?

Fossil distribution

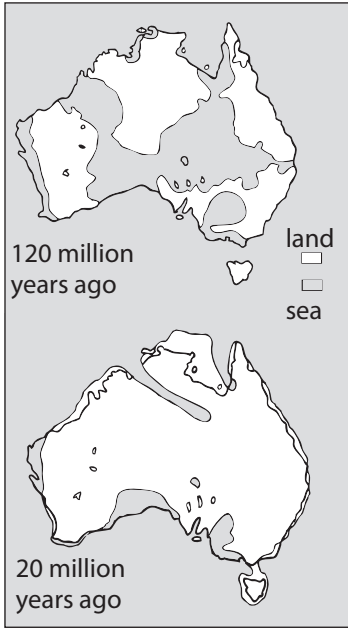
The discovery of similar fossils in different continents provided evidence for Gondwana.

- 9 Referring to figure 3, identify the fossils which were common to all continents.
- 10 Outline the similarities and differences between present-day organisms and the fossils.

Australia and sea levels

- 11 Figure 4 shows the change in distribution of land and sea surrounding Australia over time. Shade blue the regions that are land today, but were covered by water in the past.
- 12 How has the shape of the Australian land mass changed?

Figure 4 – Distribution of land and sea surrounding Australia



European settlement

European settlement has greatly influenced the Australian environment.

- 13 Complete table 1 by describing the effect of each activity on Australia’s environment – land, sea and air.

Figure 3 – Fossil distribution

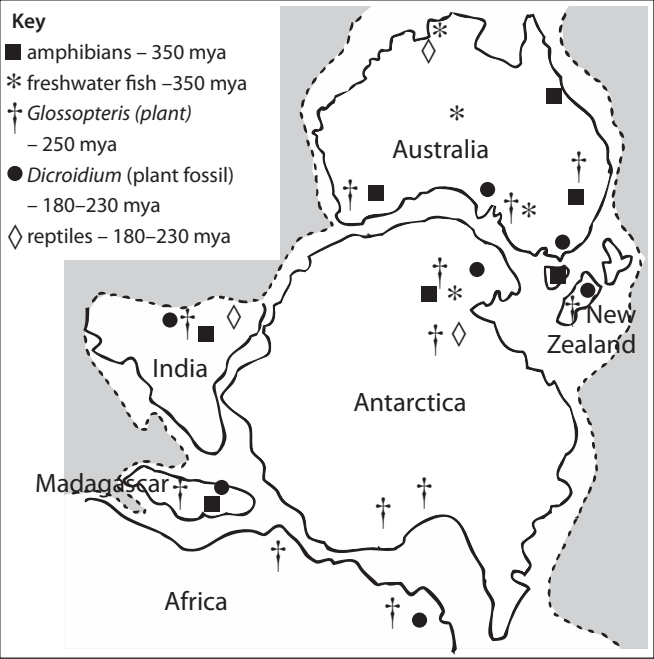


Table 1 – Effects of European settlement on Australia

Activity	Effect on the Australian environment
Urbanisation	
Mining	
Forestry and logging	
Cropping, e.g. monocultures of wheat	
Introduction of sheep and cattle and pastoral industries	
Introduction of plants and animals, e.g. cat, cane toad, rabbit, fox	
Use of chemicals, e.g. DDT	
Nuclear testing, e.g. South Australia	
Altering fire regimes	
Introducing pollutants, e.g. industrial waste	
Creating dumps and tips	
Alteration of waterways, e.g. dams and irrigation	
Tourism, e.g. Great Barrier Reef	

Extension

- * Research the evolution of plants and animals in Australia. Discuss the impact of plate tectonics, climate and humans on their survival.
- * Predict how Australia’s environment will change over the next 200 years. Include evidence and reasons for your predictions.



Alan Turing

1912–1954

Computers

- 1 Complete the information these words: *Babbage, person, engine, room, mechanical, smaller, hundred, expensive, integrated, use.*

In 1837, Charles _____ became the first _____ to conceptualise and design a fully programmable _____ computer that he called his analytical _____.

Early computers were the size of a large _____ and consumed as much power as several _____ modern personal computers.

The introduction of _____ circuit technology in the 1970s and the creation of microprocessors made computers _____, faster, more reliable and less _____.

Home computers became widely available and the introduction of the Internet and email rapidly increased their _____.

Importance and uses of computers

- 2 List 20 functions or uses of computers.

Countries with the most computers in the world

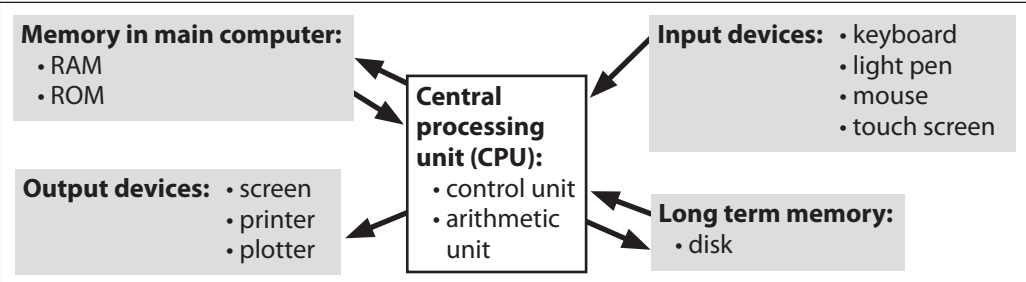
- 3 Table 1 lists the countries with the most computers in the world. Graph this data.
- 4 Which country has the greatest number of computers?
- 5 Find out the number of computers per capita for each country in table 1. What does this tell you?

Parts of a computer

A computer consists of hardware and software. The hardware is the machinery. It contains the memory and central processing unit which stores and manipulates information. The software consists of the programs that tell the computer how to process the information.

- 6 Referring to figure 1, describe the main components of a computer and their purposes.

Figure 1 – Parts of a computer



- Alan Turing was born in Paddington, London, UK. In 1934, he obtained his mathematics degree from Cambridge University and in 1938, a PhD in logic, algebra and number theory from Princeton University.
- During the World War 2, he was the chief cryptography consultant at Bletchley Park.
- From 1945–47, he worked at the National Physical Laboratory on the design of the ASCE automatic computing engine – a stored program computer
- In 1949, he became deputy director of the computing laboratory at the University of Manchester and worked on software for one of the earliest true computers, the Manchester Mark 1.
- From 1952–54, he studied mathematical biology, then died in 1954 from eating an apple laced with cyanide.
- Turing is now known as the founder of computers and computer science.

Definitions

Computer – Mechanical device with many functions, including storing and analysing data

Software – Programs of a computer

Internet – Global network of computers used for communication and commerce

Email – System used to send messages and files via the Internet

Table 1 – Countries with the most computers (2000)

Country	Number of computers
USA	164 100 000
Japan	49 900 000
Germany	30 600 000
UK	26 000 000
France	21 800 000
Italy	17 500 000
Canada	16 000 000
China	15 900 000
Australia	10 600 000
South Korea	10 600 000

Source: Computer industry Almanac Inc. Press release

Key terms

7 Write each term beside the correct meaning in table 2: *CD-ROM, DVD+RW, monitor, virus, motherboard, ROM, hard drive, RAM, CPU, modem, mouse, Internet.*

Computers in organisations and industry

- 8 Describe how and why computers are used in each area.
- a Education
 - b Gambling
 - c Banking
 - d Retail
 - e Police
 - f Travel agents
 - g Forensic science
 - h Scientists and research organisations
 - i Security
 - j Hospitals

Computers and leisure

Table 3 shows the use of computers by Australian children.

- 9 Graph the data in table 3.
- 10 Which age group uses the computer the most for the following activities?
- a email
 - b games
 - c school and education
- 11 What other trends can you see in table 3?

The Internet

The Internet is a global network that links hundreds of thousands of computer networks and millions of computers, worldwide. The Internet is based on a global array of super computers which are connected by high speed links and communication satellites.

- 12 Draw a diagram to show how the Internet works.
- 13 Why is it impossible to completely control the Internet?

Table 2 – Key terms

	Compact disk with read-only memory
	Hand-held device allowing you to scroll up and down the screen
	Modulator Demodulator – allows one computer to connect with another and transfer data between them
	Program or script that negatively affects a computer
	Random access memory
	Read-only memory data and software that is not changed, e.g. start-up programs
	Stores operating system, programs and other files
	Main circuit board containing ROM, memory expansion slots, serial ports, USB ports and hard drive
	Central processing unit
	Computer screen
	Digital versatile disk, rewritable
	Developed in 1969 by the United States military as a communications network

Table 3 – Computer use by Australian children

Activity	Age			
	5–8 yrs	9–11 yrs	12–14 yrs	All
Internet				
Email	25%	52%	69%	48%
Games	91%	86%	76%	85%
Education	70%	89%	94%	84%
Other	10%	13%	17%	13%
Frequency of use				
Every day	9%	16%	33%	19%
2–6 days/week	48%	60%	57%	55%
1 day/week	27%	8%	3%	9%

Source ABS, *Children's participation in Cultural and Leisure activities* (cat no 4901.0)

Extension

- * Outline the history and development of the computer.
- * Discuss the negative and positive benefits of technology.
- * Explain the negative aspects of computer use in these situations: viruses, hackers and security, computer addiction, violent computer games, gambling online.
- * List the advantages and disadvantages of censorship in the context of the Internet.
- * Turing was a code-breaker in World War 2, and was leader of the team that broke the Nazi's Enigma code. Find out how the Enigma machine worked and how Turing broke the code.



Fiona Wood

1958–

- Fiona Wood was born in England and was a graduate of St Thomas Hospital Medical School in London. She emigrated to Perth, Australia, in 1987.
- Fiona pioneered the use of spray-on skin cells which bound the top layer of skin together and formed a waterproof and sweat-resistant layer for 3–4 days.
- Spray-on skin technology reduced culturing time, healing time and scarring. In October 2002, her method was used to help 28 burn victims from the Bali bombings.
- Fiona is now Head of the Royal Perth Hospital Burns Unit, Professor of Paediatrics and Child Health at the University of Western Australia and Director of the McComb Research Foundation.

The importance of skin

- 1 In pairs or small groups, discuss the importance and functions of the skin. Draw a mind map to record your ideas.

Structure of skin

The skin is the body's largest organ. It protects the body, regulates temperature and is a storage site for water and fat.

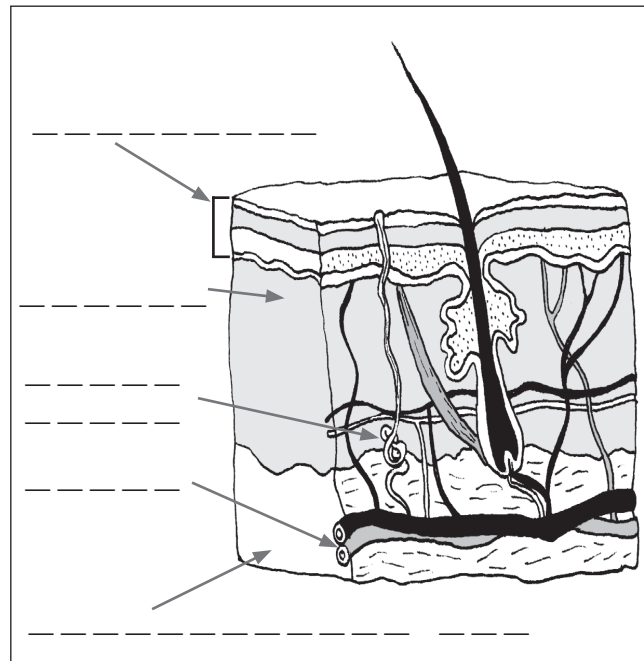
The skin is made of three main layers: the epidermis, dermis and subcutaneous fat layer.

- 2 Label figure 1 using these words: *epidermis, dermis, sweat gland, subcutaneous fat, nerve*.

Classification of burns

- 3 Complete the information using these words: *skin, burns, scalds, scarring, light, peeling, movement, electricity, chemicals, muscle*.

Figure 1 – Structure of the skin



Definitions

Burn – Tissue damage caused by agents such as sunlight, heat and chemicals

Melanoma – Form of deadly skin cancer

First degree burns – Burns that affect the outer layer of skin

Second degree burns – Burns that damage the epidermis and dermis region of the skin

Third degree burns – Burns that damage all layers of the skin

First degree _____ affect the outermost layer of the _____, called the epidermis. They are often caused by UV _____ or contact with hot objects. They result in pain, redness, swelling and _____.

Second degree burns affect the second layer of skin, called the dermis. They are often caused by UV exposure and result in blisters and _____. Blisters can take 2–6 weeks to heal and _____ can occur.

Third degree burns affect the epidermis, dermis and hypodermis. They are often caused by _____ or corrosive _____. _____ and nerve endings are damaged, and scarring and tightening can occur, which prevents normal _____.

- 4 Draw and label diagrams to show first, second and third degree burns.

First degree burns

Second degree burns

Third degree burns

- 5 Young children have thinner skin than older children. How could this affect the degree of burns they receive?

Causes of burns

6 Complete table 1 by describing a situation in which burning could occur, for each source. Then provide safety tips for avoiding burns.

Treatment

First degree burns should be covered with cool, moist compressions to reduce pain and swelling.

Second degree burns require cooling, washing with saline water and consultation with a doctor.

Third degree burns require emergency medical services.

- 7 Draw and label diagrams to show the treatment that should be given for each type of burn.
- 8 Why should oils such as butter and margarine never be applied to burns?

Preventing burns

- 9 Explain how each of the following could reduce the incidence of burns.
- a Smoke detectors
 - b Flammability regulations on clothing
 - c Open fires replaced with electrical heating
 - d Work safety, e.g. fire drills and fire extinguishers
 - e Elimination of cigarette smoking in the community
 - f Water heaters set to lower temperatures

Sun burn, skin cancers and melanomas

Melanomas are the fastest growing cancer in the world – 160 000 new cases are diagnosed annually. Australia, New Zealand, North America and Northern and Western Europe have the highest incidence rates.

Advanced melanomas cannot be cured. The average survival time after diagnosis is only seven months.

- 10 Why is there a high incidence of melanomas in Australia, New Zealand, North America and Northern and Western Europe?

Table 2 shows the incidence of melanomas per 100 000 population in the world per year.

- 11 Graph this data, and describe the shape of your graph.
- 12 Suggest reasons why the incidence of melanomas has been increasing rapidly?
- 13 How can people reduce their exposure to the sun in order to avoid melanomas?

Table 1 – Causes of burns

Source	A situation where burning could occur	Safety tip for how this could be avoided
Fire		
Heat		
Electricity		
Radiation		
Chemicals		
Hot objects, e.g. cooking appliances		
Hot liquids		
Bathtubs		
Barbecues		
Fireworks		

Table 2 – Incidence of melanomas

Country	Incidence per 100 000 of the population per year
Australia	29
New Zealand	27
USA	12
Denmark	10
UK	6
Italy	4

Source: Parkin D.M, Whelan S.L, Ferlay J, Raymond L and Young J eds (1997) *Cancer in Five Continents Vol II* (IARC Publications No 143) Lyon, IARC.

Extension

- * Research any links between the state of the ozone layer and the incidence of melanoma.
- * Skin cancer is a preventable disease. Discuss what strategies could reduce the incidence of skin cancer, worldwide?
- * *Lifestyle factors are the key cause of increasing cancer rates in the world.* Discuss this statement.



Robert Jarvik

1946–

- Robert Jarvik was born in Midland, Michigan, USA. He obtained a medical degree from Syracuse and New York University and was inspired to build an artificial heart after his father died of heart disease.
- In 1982, he developed the Jarvik 7, which provided patients with a temporary heart until a transplant was available.
- He founded Jarvik research in 1987 and developed the Jarvik 2000, which was a thumb-sized pump that helped alleviate the strain on the heart.
- Today he works with leading cardiologists and cardiac surgeons.

The heart

The heart is a muscular organ which pumps blood around the body.

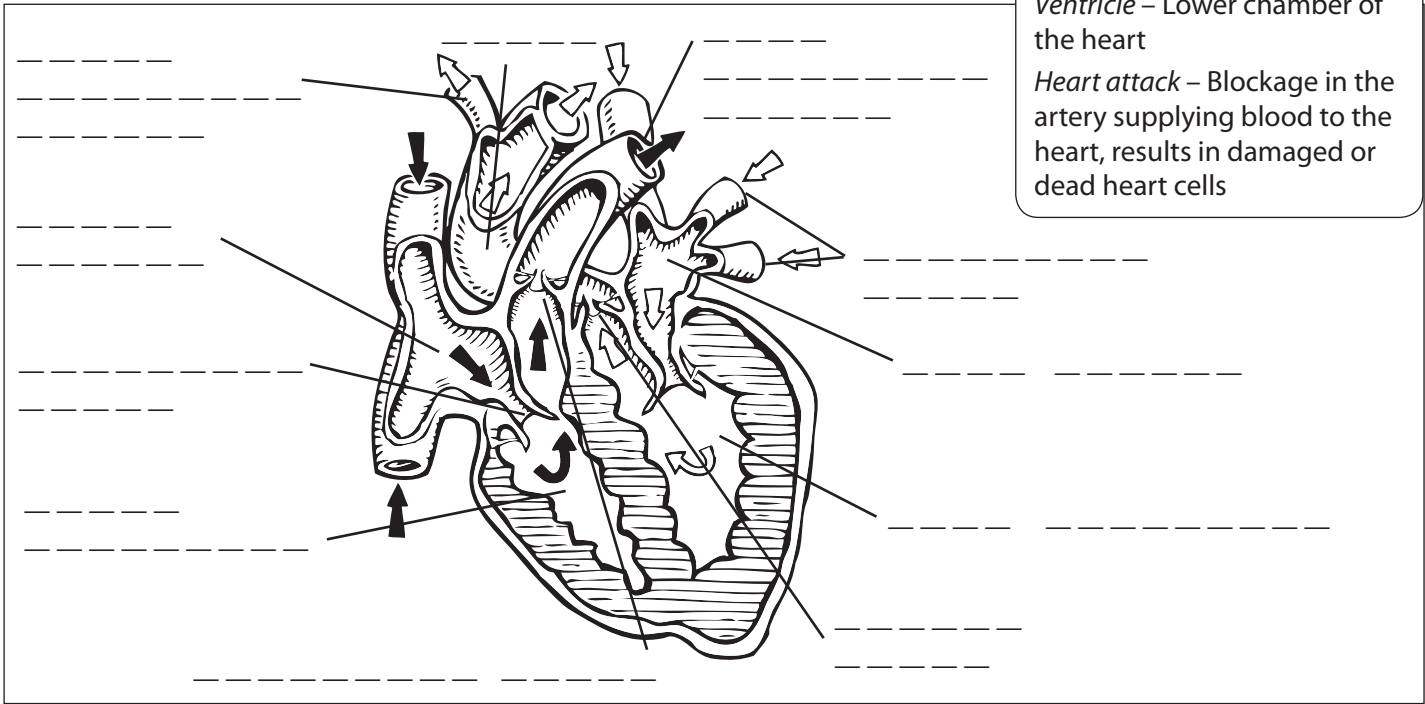
It consists of four chambers – the upper atria and lower ventricles.

The heart pumps 100 000 times each day and 2.5 billion times in an average lifetime. It moves 8 tonnes of blood every 24 hours.

Structure of the heart

- 1 Label the heart in figure 1 using these words: *aorta, left pulmonary artery, pulmonary vein, left atrium, left ventricle, aortic valve, pulmonary valve, right ventricle, tricuspid valve, right atrium, right pulmonary artery.*

Figure 1 – Structure of the heart



Definitions

- Heart* – Muscular pump that transports blood around the body
- Jarvik 7* – Artificial heart developed by Robert Jarvik
- Atrium* – Upper chamber of the heart
- Ventricle* – Lower chamber of the heart
- Heart attack* – Blockage in the artery supplying blood to the heart, results in damaged or dead heart cells

Importance of the heart

- 2 Complete the information using these words: *million, disease, coronary, leading, attack, burden, strokes, arteries, oxygen.*

Cardiovascular _____ is the cause of 16.7 _____ or 29.2% of deaths worldwide. By 2010, cardiovascular disease will be the _____ cause of death in developing countries.

Cardiovascular disease refers to heart and blood _____ disease, and can lead to angina and heart _____. Heart attacks are a result of a blockage in one of the arteries supplying blood to the heart. A stroke occurs when there is a blockage in one of the _____ in the brain, and 80% are caused by a blood clot in the brain and 20% by a blood vessel bursting. In both cases, the brain is starved of oxygen and cells are damaged or killed.

Coronary heart disease occurs when the arteries that supply the heart with oxygen become clogged and insufficient _____ reaches the heart.

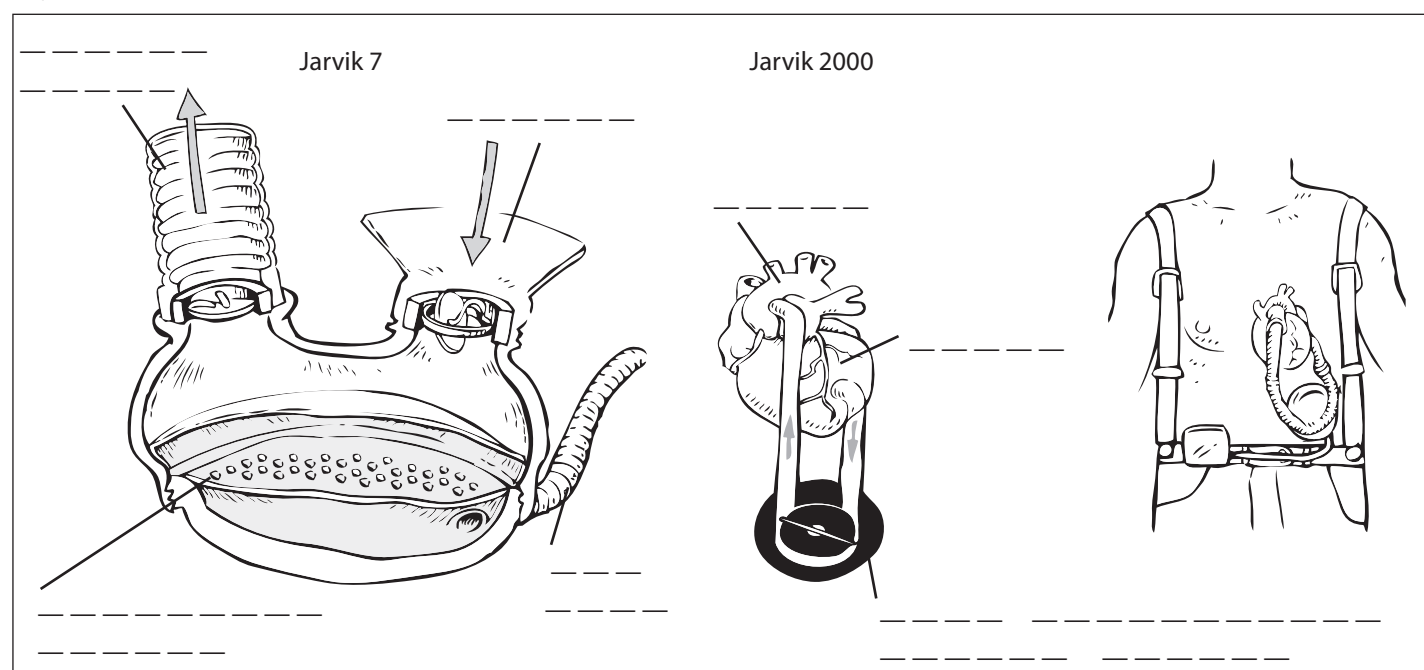
20 million people survive heart attacks and _____ every year and require costly clinical care. The health and economic _____ of cardiovascular disease exceeds that of any other disease.

Temporary treatment of heart disease

The Jarvik 7 and Jarvik 2000 hearts are devices which can keep the patient alive until a transplant is available. The Jarvik 2000 is a miniature rotary pump and is electrically powered.

- 3 Label the parts of the Jarvik hearts using these words: *airline, mitral, aortic valve, diaphragm layers, heart, left ventricular assist device, aorta*.

Figure 2 – Structure of the Jarvik 7 and Jarvik 2000



- 4 For each part of the Jarvik hearts you labelled on figure 2, briefly describe its function or purpose.
- 5 Why is there a shortage of hearts available for transplants?
- 6 Suggest reasons why few people donate their organs.

Diagnosis and treatment of heart problems and related conditions

- 7 Write each term beside the correct meaning: *heart transplant, pacemaker, angiography, bypass surgery, defibrillator, blood pressure, electrocardiogram, medication*.

a Special X-ray of the heart and coronary arteries

b Drugs used to treat angina and high blood pressure, e.g. aspirin

c Bypass of the narrow areas in coronary arteries to allow blood to reach the heart muscle

d Small electrical device used to stimulate the heart to beat

e The force exerted in the arteries by blood as it circulates.

f Machine used in emergencies and cardiac arrest to give the heart an electric shock and activate it

g Removal and replacement of diseased heart

h Reading of the heart's electrical impulses

Extension

- * Discuss the social, cultural and economic factors that contribute to heart disease and how its incidence can be reduced.
- * Find out what the symptoms of heart disease, angina, heart attacks and strokes are.
- * Design a brochure or PowerPoint presentation to promote a lifestyle that reduces risks of heart disease.



Martin Cooper

1928–

- Martin Cooper was born in Chicago USA. He obtained a degree in Electrical Engineering from the Illinois Institute of Technology.
- Martin then spent four years in the navy on destroyers and submarines.
- In 1954, he joined Motorola and helped develop the first hand-held radios for the Chicago police department.
- Martin became the General Manager of the Systems Division at Motorola and in 1973 developed the mobile phone.
- More than two billion mobile phones are now used worldwide.

Importance of mobile phones

- 1 Complete the information using these words: *mobile, network, radio, two, electronic, rate, contracts, communication, tasks*.
- A _____ phone is a portable _____ device used for telecommunications over long distances. The phones are connected to a _____ of base stations, which relay messages using _____ and microwaves.
- More than _____ billion mobile phones are used worldwide, and Africa has the largest increase in growth _____ in the world. The availability of prepaid or pay-as-you-go services and the lack of long term _____ has helped fuel this growth.
- Mobile phones are a quick, simple and efficient form of _____ and can perform a multitude of _____.

Definitions

- Mobile phone* – Portable device used for telecommunications. Electromagnetic waves and base stations are used to relay information
- Electromagnetic* – Electric and magnetic fields that travel at the speed of light
- Wave* – e.g. Light, radio and microwaves
- Base station* – Antenna which relays mobile phone messages
- Recycling* – To reuse mobile phones to minimise waste and environmental contamination

The first mobile phone

The characteristics of the first mobile phone are shown in table 1.

- 2 How much did the mobile phone weigh?
- 3 For how long could you speak before the phone had to be recharged?
- 4 How long did it take to recharge the mobile phone?
- 5 Fill in the last column with comparable data for a modern mobile phone. Then write a short paragraph outlining the similarities and differences,

Table 1 – Characteristics of the first mobile phone

Name	Motorola Dyna-Tac	
Size	22.86 x 13 x 4.5 cm	
Weight	1.2 kg	
Display	None	
Number of circuit boards	30	
Talk time	35 minutes	
Recharge time	10 hours	
Features	Talk, listen and dial	

Modern mobile phones

- 6 Draw a diagram of a modern mobile phone and illustrate the features that it contains. Annotate your diagram to describe how technology and user needs have impacted on phone design.

Use of mobile phones

Table 2 shows the percentages of households in Australia with mobile phones from 1998–2004.

- 7 Graph the data in table 2.
- 8 Describe the shape of your graph. What does it tell you?

Table 2 – Percentages of Australian households with a mobile phone, 1998–2004

Year	1998	2000	2002	2004
Households with a mobile phone	44%	61%	72%	85%

Source: Australian Bureau of Statistics (ABS) Household use of information technology (cat number 8146.0) ozTAM,DBA

9 What factors have contributed to the popularity of mobile phones?

Negative effects of mobile phones

Mobile phone etiquette

10 For each situation in table 3, list a specific example of how a mobile phone can be a problem, then explain the reason it is a problem.

Table 3 – Mobile phone etiquette

Situation	Example	Reason a problem
Schools		
Funerals		
Cinema and theatre		
Churches		
Photographing people without consent		
Driving while using a mobile phone		
Using a mobile phone to bully and intimidate people		
High debt accumulation		

Brain cancer

Mobile phones and their base stations receive and emit radio and microwaves. Many consumers are concerned about a link between mobile phones, brain tumours and DNA damage. Research has been inconclusive. However, engineers who work on high power radio transmitters have a greater incidence of premature cataracts. The World Health Organisation has introduced the precautionary principle regarding mobile phones. This is a form of risk management which is used when there is scientific uncertainty.

- 11 Describe how each management practice listed below could minimise the risk from mobile phones.
- a Minimise use of phone

b Limit use by children

c Use hands-free technology

d Purchase a phone with the lowest radiation emission levels

e Do not locate base stations in high population areas, such as schools

Recycling mobile phones

On average, people replace their mobile phones every 18 months. Fifteen million are discarded in the United Kingdom each year, but only 4% are recycled. Ten of the most dangerous substances on Earth are used in mobile phones, including cadmium, rhodium, palladium, beryllium, arsenic and mercury. Many mobile phones are dumped in land fills and their contents leak into the surrounding soil.

- 12 Suggest why so many mobile phones are discarded, and few recycled.
- 13 Discuss the environmental consequences of discarded mobile phones.

Extension

- * Find out what mobile phone recycling programs there are in your local area.
- * Conduct a survey around your school or community to find out how many mobile phones are being discarded, and how many recycled.
- * Discuss the benefits of recycling or refurbishing mobile phones for use in disadvantaged regions.
- * All mobile phones should be banned until the potential risks of use have been fully researched. List arguments for and against this statement.
- * Discuss the positive and negative effects of mobile phones on our society.
- * Research and construct a timeline showing the history of devices that people have used for communication.



Fiona Stanley

1946–

- Fiona Stanley was born in Sydney, Australia.
- In 1970, she graduated in medicine from the University of Western Australia and undertook further studies in the United Kingdom and United States of America.
- Fiona researched children's health problems. In 1995, she found that taking folic acid during pregnancy could prevent spina bifida.

Spina bifida

- 1 Complete the information using these words: *birth, incidence, spina, nervous, conception, family, environmental, folic, ultrasounds*.

The term '_____ bifida' is derived from the Latin *spina* meaning 'spine' and *bifida* meaning 'split or divided'.

Spina bifida is a serious _____ defect caused by the incomplete formation of the brain, spinal cord and _____ tissues.

Initially, 2 in every 1000 children were affected by the condition, and it was believed to be caused by genetic and _____ factors.

If there is a close _____ history of the disease, there is 1 chance in 25 of having a child with spina bifida.

_____ acid taken one month before _____ and for the first three months of pregnancy can reduce the risk of having a child with spina bifida by 70%.

The deformity can be detected by _____, where the spine appears to have a soft lump or lesions. Surgery to place the nerves and tissues under the skin must take place within 24 hours of birth.

Since 1996, education programs have helped reduce the _____ of spina bifida to 1.3 children per 1000 births.

- 2 Define the term *spina bifida*.
- 3 Outline the causes of spina bifida.
- 4 At what stage during pregnancy can spina bifida occur?

Problems associated with spina bifida

- 5 Label figure 1 to show the problems associated with spina bifida.

Problems associated with spina bifida

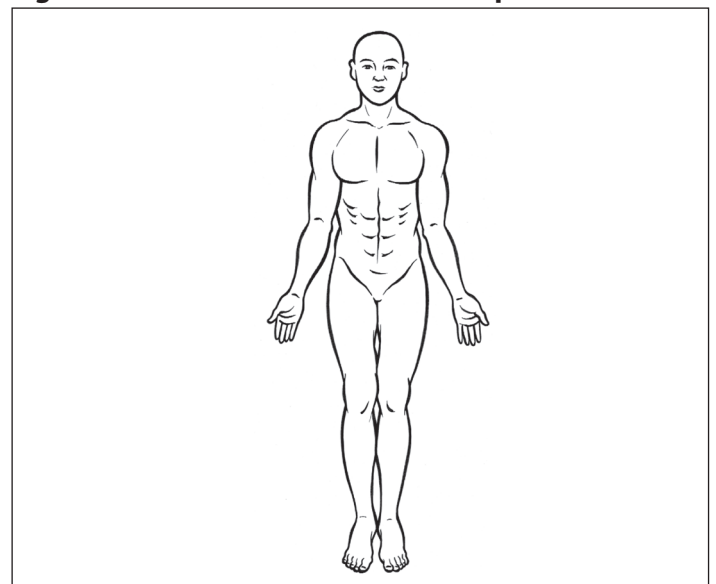
- * Paralysis
- * Muscle weakness
- * Reduced feeling and sensation
- * Difficulty walking
- * Proneness to burns and pressure sores
- * Bowel and bladder incontinence
- * Learning problems
- * Accumulation of fluid on the brain with associated brain damage, seizures or blindness
- * Limb malformations
- * Deformities of the spine
- * Needs aids to move e.g. crutches, braces or wheel chair

Definitions

Spina bifida – A deformity caused by the incomplete closure of the spinal column

Folate/Folic acid – Water-soluble vitamin found in fruit and vegetables

Figure 1 – Problems associated with spina bifida



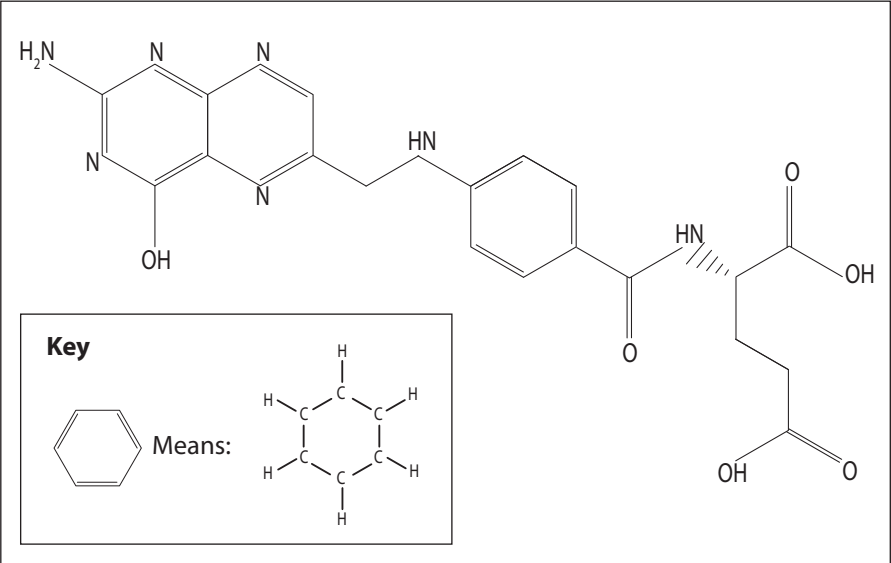
Importance of folate

Folate is a water-soluble vitamin found in grains, meat, fruit and vegetables. Low levels have been associated with increased risk of miscarriages, premature delivery and low infant birth weight. The recommended daily allowance for folate is 400 mcg per day for women.

Insufficient folate can have an adverse effect on:

- * formation of red blood cells
 - * formation of genetic material
 - * protein metabolism
 - * cell growth and division
 - * the development of the spinal column.
- 6 Referring to figure 2, outline the key chemical elements that make up folic acid.
- 7 Describe a diet that could be deficient in folic acid. Then outline how that diet could be improved.

Figure 2 – Folic acid



Education and spina bifida

Table 1 shows the incidence of spina bifida from 1990–2000.

- 8 Graph the data in table 1.
- 9 Describe the trends in the data.
- 10 Discuss the effect of research and education on the incidence of spina bifida.

Table 1 – Spina bifida per 1000 births, 1992–2000

Year	Rate
1992	2.5
1993	2.3
1994	2.5
1995	2.6
1996	2.7
1997	2.5
1998	2.4
1999	2.1
2000	2.1

Source: National Centre for Health Statistics

Diet and disease

- 11 List ten diseases caused by diet, e.g. rickets. For each one, explain how the diet could be modified to reduce the likelihood of developing the disease.
- 12 Describe the conditions anorexia and bulimia. Why are they a problem in developed countries?
- 13 How has social, cultural and technological change affected our nutrition?

Technology and food production

- 14 Complete table 2 by outlining the effect of technology on the food that we eat.

Table 2 – Technology and food production

Technology	Effect on food
Refrigeration	
Microwave oven	
Preservatives and additives	
Sterilisation	
Genetic engineering	
Feedlots and intensive production	
Chemicals, e.g. herbicides and pesticides	
Processing and packaging	

Extension

- * Research birth defects that have been identified by science, and for which causes have been found.
- * Discuss the causes and effects of obesity in our society.
- * *Anorexia nervosa is a psychological condition rather than a diet-related problem.* Discuss this statement, and the impact of the disease on those who suffer from the condition, and their families and communities.



John Tyndall

1820–1893

- John Tyndall was born in Ireland. In 1839, he joined the Irish Ordnance Survey and became a draftsman and surveyor.
- He taught maths at Queens Wood College and in 1848 undertook a PhD in magnetism.
- In 1853, John was appointed Professor of Natural Philosophy at the Royal Institution.
- In 1870, he demonstrated that light follows the curve of a stream of water pouring from a container. He also invented the light pipe, which led to the development of fibre optics.

Fibre optics

1 Complete the following paragraph using these words: *metre, light, fibres, millions, laser, telecommunications*.

Optical _____ are long, thin, transparent strands of glass which transmit pulses of _____ generated by a laser.

The pulses of light are produced _____ of times per second and reflect several times per _____ along the walls of the fibre.

Optical fibres are used in many industries, including _____ and medicine, and in lighting, security and industrial _____ systems.

Definitions

Optical fibres – Thin glass tube through which light travels

Laser – Machine that creates pulses of light

Internal reflection – Light changes direction, and angle of incident light equals angle of reflected light

Structure of fibre optics

A fibre optic consists of a glass core, a denser glass cladding and a buffer coating.

Light is reflected as it travels along the optical fibre. This occurs because the walls of the fibre have different densities.

2 Complete figure 1 by showing the movement of light as it reflects along the optical fibre.

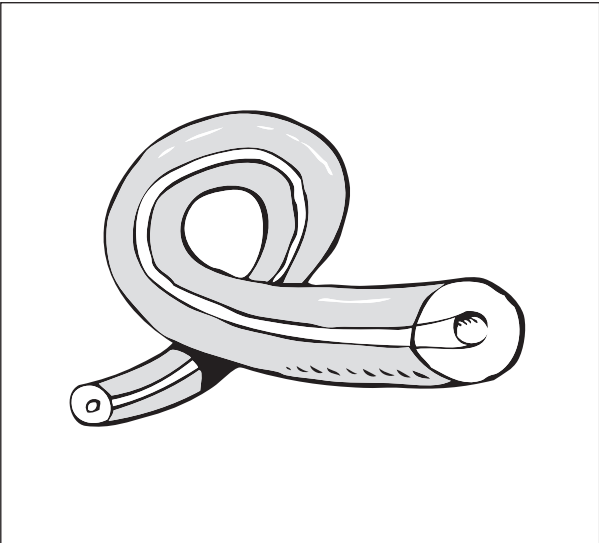
Uses of fibre optics

3 Write each use of optical fibres beside the correct example in table 1: *medical, defence, telecommunications, traffic control, security, sensors, lighting, industrial*.

Table 1 – Uses of optical fibres

Use	Examples
	Illumination inside buildings
	Television, telephones and computers
	Detect changes in temperature, pressure and chemical concentrations
	Closed circuit television
	Keyhole surgery, endoscopic investigations
	Transport lasers to distant sites for welding, cutting or drilling
	Wiring in aircraft and submarines
	Automated toll booths

Figure 1 – Transmission of light in an optical fibre



4 Complete figure 2 by listing 10 devices you use every day that utilise fibre optics.

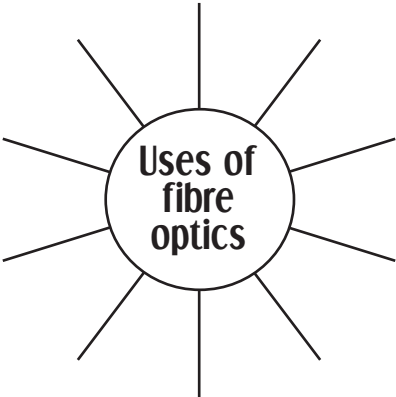


Figure 2 – Uses of fibre optics

Advantages of fibre optics

- 5 For each characteristic of fibre optics below, list three specific applications or situations in which fibre optics are an advantage.
- a Lightweight
 - b Small
 - c Flexible
 - d Can be bundled into cables
 - e Not affected by electromagnetic radiation
 - f Interference, e.g. radio signals and lightning
 - g Large bandwidth
 - h Safe in explosive regions, as glass is not an electrical conductor and a spark cannot trigger an explosion
 - i Greater security (cannot tap into a fibre cable to read the data signals of banks and security installations)
 - j Use less energy than copper systems

Disadvantages of fibre optics

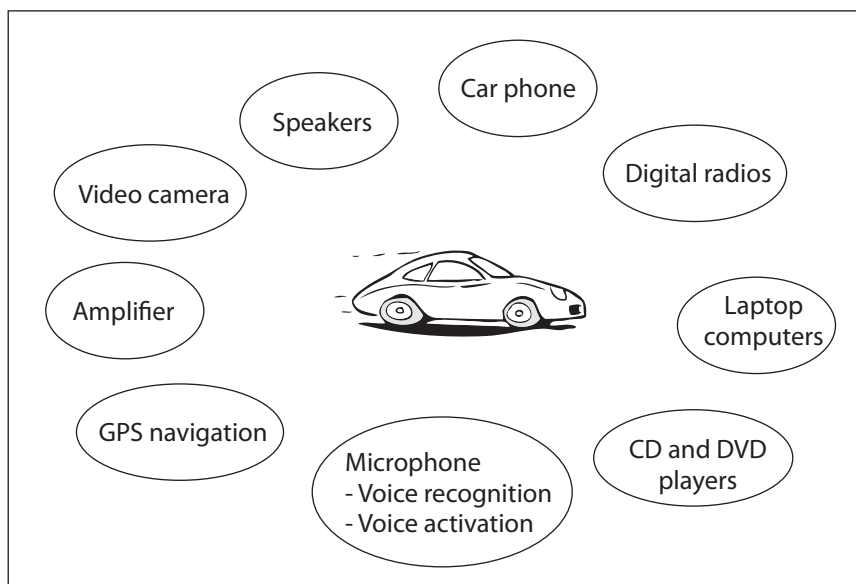
- 6 Explain, using real-world examples, how each characteristic could be a disadvantage of fibre optics.
- a More expensive than copper
 - b Difficult to connect
 - c Specific equipment required
 - d Specialist training required to use fibre optics

Cars

Figure 3 shows the systems controlled by fibre optics in a car.

- 7 Write a general statement about the types of systems controlled by optical fibres in a car.
- 8 List the advantages and disadvantages of using fibre optics in a car, and give reasons for your answers.
- 9 List three uses of fibre optics in cars that you think might be developed in the future.

Figure 3 – Uses of optical fibres in cars



Security

- 10 Explain why security is critical for institutions that use optical fibre systems, such as banks.

True or false?

- 11 Mark each statement as true or false. Then correct the false statements so that they are true.
- a Optical fibres transmit information using light
 - b Optical fibres cannot be placed on ocean floors
 - c Optical fibres are cheaper than traditional cables
 - d Optical fibres have greater security than traditional systems
 - e Optical fibres are affected by electromagnetic radiation
 - f Optical fibres are not as durable as copper cable

Extension

- * Research the medical applications of fibre optics.
- * Discuss the use of optical fibres in the telecommunications industry.
- * How do you think optical fibres will be used in our society in the future?



Wallace Carothers

1896–1937

- Wallace Carothers was born in Iowa, USA. He obtained his Science degree from Tarkio College, Missouri, in 1920 and his PhD from the University of Illinois in 1924.
- In 1928, he started work with the chemical company DuPont. Six years later he discovered the synthetic polymer nylon.
- Wallace published 31 papers and established general theories of polymers.
- He is believed to have suffered from bipolar disorder and committed suicide in 1937 after the death of his sister.

Nylon

- 1 Complete the information using these words: *parachutes, demand, coal, nylon, polymer, stockings, silk, 1941, production.*

_____ refers to a family of synthetic thermoplastics. It is made from _____, water and air. It is a _____, which means it consists of repeating units.

Nylon was developed to replace _____. It was first marketed in 1938 when it was used in toothbrushes and _____.

Nylon was used for the mass production of _____ after the United States entered World War 2 in _____.

Nylon 6 and nylon 66 account for approximately 98% of total nylon _____.

Nylon is used in fabrics, ropes, clothing, carpets, packaging, mechanical parts, parachutes and tennis racquets.

Global _____ for nylon fibres is more than 4 million metric tonnes per year.

Definitions

Nylon – Synthetic chemical used to make stockings, rope and many other items

Polymer – Chemical made up of repeating units

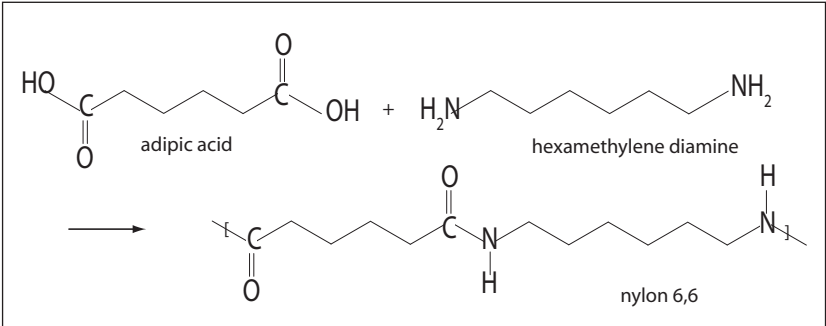
Silk – Natural fibre made by silkworms

Making nylon

Nylon is made by reacting adipic acids with hexa methylene diamine.

- 2 Referring to figure 1, what are the elements in nylon?
- 3 Explain why nylon is classified as a polymer.

Figure 1 – Production of nylon 66



Advantages of nylon

- 4 Complete table 1 by outlining the advantages and applications for each characteristic of nylon.

Table 1– Characteristics of nylon

Characteristics	Advantage	Used in
Lightweight	easy to carry	mountain climbing ropes
Strong		
Durable		
Elastic		
Easy to wash		
Dries quickly		
Retains its shape		
Can be permanently set by heat		
Can be dyed in a large range of colours		
Smooth and soft yarns		

Uses of nylon

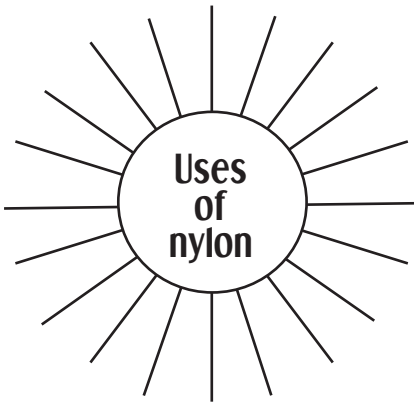
Figure 2 – Uses of nylon

5 Complete figure 2 by listing 20 uses of nylon.

Suitability for different uses

6 Outline the features of nylon that make it suitable for these products.

- a Stockings
- b Swimwear
- c Parachutes
- d Tyres
- e Life vests
- f Carpets
- g Seat belts
- h Ropes
- i Tents



Disadvantages

- 7 Nylon absorbs little water. Discuss how this could be a disadvantage when designing sports clothing?
- 8 Why are babies' clothes made of cotton mixtures rather than pure nylon?

Tensile strength

Rope was traditionally made from the hemp plant, and its construction required no industrial processes. However, damage and wear were difficult to detect because it rotted from the inside.

Many ropes are now made of nylon because it is twice as strong as hemp and weighs 25% less.

Table 2 lists the percentage strength of nylon safety equipment after several years of service.

- 9 Graph the data in table 2.
- 10 What trends can you see?
- 11 What implications does this data have for safety equipment? Consider examples such as life rafts, which are continually exposed to the environment.

Table 2 – Effect of age on the tensile strength of nylon safety equipment

Age of equipment (years)	Tensile strength %
0	100
1	95
2	90
3	80
4	70
5	65

True or false?

- 12 Mark each statement as true or false. Then correct the false statements so that they are true.
 - a Nylon is a synthetic chemical.
 - b Nylon is highly flammable.
 - c Nylon has replaced silk.
 - d Nylon is derived from fossil fuels.
 - e Nylon is stronger than natural fibres.
 - f Nylon is a polymer.
 - g Nylon is made up of the elements hydrogen, helium, oxygen, carbon and nitrogen.

Extension

- * Discuss the use of nylon in our society. Outline its advantages and disadvantages.
- * *Non-renewable fossil fuels should not be used to make fibres when natural fibres are readily available.* Discuss this statement.
- * *Natural fibres will always be used because of their superior characteristics.* Discuss this statement.
- * Research three other synthetic polymers that have been developed in the last 100 years.



Sir Howard Florey

1898–1968

- Howard Florey was born in Adelaide, Australia. He studied medicine and pathology then undertook research at Cambridge and Sheffield University.
- In 1928, Alexander Fleming observed that certain moulds inhibited the growth of bacteria.
- Howard and Ernest Boris Chain isolated and purified this mould, penicillin, and the three men were awarded the Nobel Prize in 1945 for their discovery.
- Howard was Professor of Pathology at Oxford University, 1935–62.
- Fifty million kilograms of penicillin are now produced each year, and it has been the most widely used antibiotic in history.

Penicillin

- 1 Complete the information using these words: *fungus, antibiotics, viruses, pneumonia, wall, allergic*.

Penicillin refers to a group of _____ derived from a mould.

It is produced by _____ of the genus *Penicillium* and is effective against diseases such as syphilis, meningococcal meningitis, gangrene, _____ and some staphylococcal and streptococcal infections.

Protozoans, _____ and most fungi are not affected by penicillin.

Penicillin kills bacteria by interfering with cell _____ production.

It can cause _____ reactions, including diarrhoea, vomiting, pain, inflammation and allergic shock.

Definitions

Penicillin – An antibiotic produced by fungus of the genus *Penicillium*

Resistance – When a disease is not affected by a treatment

Antibiotic – Medication used to kill or stop the growth of a disease-producing organism

Chemical structure of penicillin

- 2 Figure 1 shows the chemical structure of penicillin. What elements does it contain?

Penicillin production

The fungus *Penicillium chrysogenum* is grown in fermentation tanks with inoculum hyphae and a growth medium of corn steep liquor, glucose, nitrates and phenyl ethanoic acid.

After 160–200 hours of fermentation, the broth is filtered and the penicillin is extracted.

- 3 Suggest why temperature and pH are carefully regulated during penicillin production.

True or false?

- 4 Answer true or false to the following statements, and explain your reasoning.

a Penicillin is derived from moulds.

b Penicillin can control all microbial infections.

c Many soldiers in World War 2 would have died without penicillin.

d Penicillin stops bacteria from making cell walls when they reproduce.

e Howard Florey discovered penicillin.

f Resistance to penicillin will be a greater problem in the future.

Resistance

Resistance occurs when a microbe is not affected by an antibiotic. Resistance to penicillin was first observed in 1947. Table 1 shows the resistance of pneumococcal infections to penicillin over time.

- 5 Graph the data in table 1.

Figure 1 – Chemical structure of penicillin

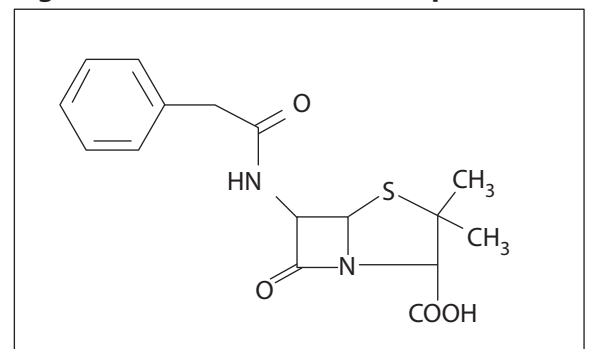


Table 1 – Resistance of pneumococcal infections to penicillin

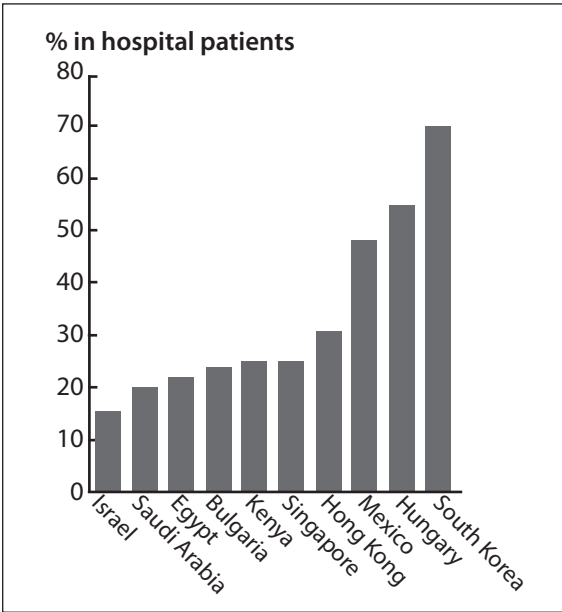
Year	% Resistance to penicillin
1997	9.6
1998	13.7
1999	26.4
2000	34.6

- 6 Describe the shape of your graph. What trends can you see?
- 7 Outline how resistance develops.
- 8 Explain why antibiotics should not be over-prescribed to the public.
- 9 Why is it essential to complete a prescription of medication, rather than stop when you feel better?

Figure 2 shows the resistance of pneumococci in different parts of the world during the 1990s.

- 10 What was the percentage of patients affected in these countries?
 - a Kenya
 - b Hong Kong
 - c Mexico
 - d Hungary
- 11 Why does the resistance level vary between regions?
- 12 What factors could have contributed to these high resistance levels?
- 13 What problems will be faced in the future as more microbes become resistant to modern drugs?

Figure 2 – Resistance of pneumococci to penicillin, 1990s



Other causes of disease

- 14 Bacteria is not the only cause of disease. Complete table 2 by listing examples and diseases for each cause.

Table 2 – Causes of disease

Cause of disease	Example of specific cause	Disease resulting
Viral		
Fungal		
Parasites		
Environmental factors		
Heredity		

Disease control

- 15 Outline how each factor can help to control the spread of disease.
 - a Quarantine
 - b Vaccination
 - c Education
 - d Medication
 - e Clean water and sanitation

Extension

- * Discuss the advantages and disadvantages of modern practices used to control disease.
- * Research five diseases and discuss their symptoms, effects and methods of control.
- * Research ‘superbugs’, including how scientists are working to resolve this problem, both in the short term and the long term.



Fred and Gabi Hollows

- Fred Hollows was born in New Zealand in 1929 and became an eye surgeon. Gabi was born in 1953 and met Fred while she was training to become an orthoptist.
- The Hollows launched a national program in the 1970s to control trachoma and other eye diseases in Indigenous communities.
- In three years, they treated 40 000 people, performed 1000 operations and prescribed more than 10 000 pairs of glasses.
- Later, they established eye programs and lens factories worldwide.
- Today, the Fred Hollows Foundation continues community-based work with blindness prevention, worldwide.

Structure of the eye

- 1 Complete the information using these words: *cornea, pupil, eye, iris, light, tissue, rods, cones, optic nerve.*

The _____ is a sense organ which responds to _____. It is enclosed by the socket and eyelid and contains nervous _____ which is stimulated by light.

The _____ is a transparent cover at the front of the eye and helps refract light. The _____ is the coloured ring of muscle and has a central opening called the _____. Light travels through the pupil and lens to the retina, which is a light-sensitive layer at the back of the eye.

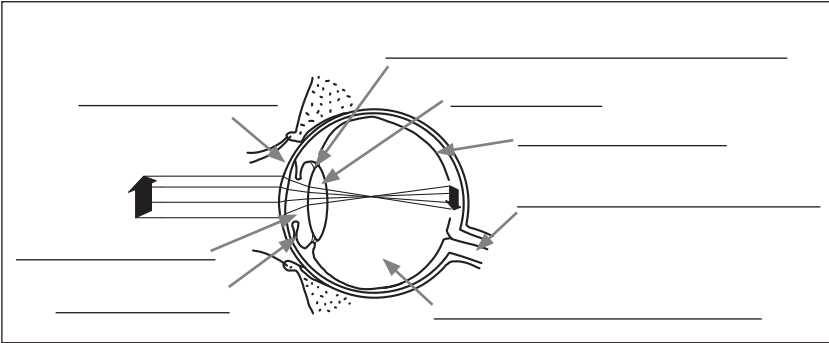
The retina consists of _____ and _____, pigmented cells and nerves. Light stimulates the rods and cones, and nerve cells send impulses via the _____ to the brain.

- 2 Label figure 1, using the information from question 1 and these words: *cornea, pupil, iris, lens ligament, lens, vitreous gel, retina.*

Definitions

- Orthoptist* – Eye specialist
Refract – To bend, e.g. light
Transmission – To spread, e.g. a disease
Inverted – To be upside-down
Trachoma – Bacterial disease of the eye that can result in blindness

Figure 1 – Structure of the eye



Blindness

Trachoma

- 3 Complete the following statements to find out more about trachoma. Use these words: *Africa, Asia, pain, eyelids, flies, gnats, blindness, 2700 AD, 400 million, adulthood, Chlamydia trachomatis, America, Islands, antibiotics.*

Trachoma is a highly contagious disease and is a major cause of _____, worldwide.

The incidence of trachoma was first recorded in _____. It affects more than _____ people and is common in North and Sub Saharan _____, the Middle East, Southern _____ and China. Pockets of the disease also occur in Latin _____, Australia and the Pacific _____.

Trachoma is a preventable disease and is caused by the bacteria _____. Infection often takes place during childhood. However, blindness usually occurs in _____.

Symptoms are similar to conjunctivitis and include: _____, swelling of the _____, inward turning and scarring of the upper eyelids, rubbing of the cornea, ulcers and blindness at 40–50 years of age.

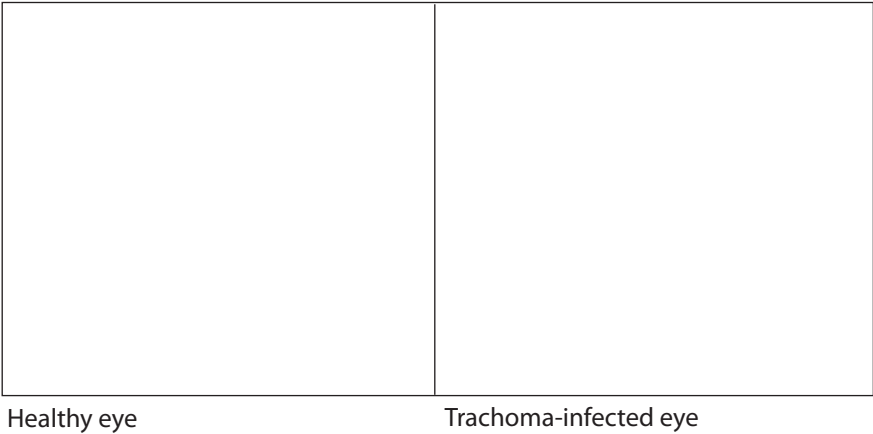
The disease is spread by direct contact with discharges from the eye or from contaminated hands, clothing, _____ and _____.

Prevention includes education, sanitation and minimising overcrowding and poor housing conditions.

The four main components of controlling trachoma include surgery, use of _____, facial cleanliness and environmental changes.

- 4 Draw and label diagrams in figure 2 to show the symptoms and differences between a healthy eye and a trachoma-infected eye.
- 5 What regions of the eye are damaged by trachoma? How does this affect the ability of the eye to focus and see images?
- 6 Complete figure 3 by drawing and annotating diagrams to show the cause, method of transmission and control of trachoma.

Figure 2 – Healthy and trachoma-infected eye



Healthy eye

Trachoma-infected eye

Figure 3 – Cause, transmission and control of trachoma

Cause	Transmission	Control

Methods of controlling trachoma

- 7 Write each method of control beside the correct description in table 1: *medicine, sanitation, education, surgery, hygiene, adequate diet, .*
- 8 Explain why trachoma has not been eradicated in some regions.
- 9 Outline the social and economic implications of blindness, for both the individual affected and their community.
- 10 Read the following paragraph and describe the trends that are apparent.

Table 1 – Controlling trachoma

Method	Description
	Medical operation to reverse in-turned eyelids
	Cleanliness, especially the face
	Clean water supplies, sewerage and rubbish removal
	Training and advice
	Food and nutrition
	Antibiotics, e.g. tetracycline drugs or sulfonamides

The percentage of people in Australia that are actually blind because of trachoma is 0.2% in the non-Indigenous community and 1.5% in the Indigenous population.

Further, the percentage of the Indigenous population that are blind from trachoma at 60 years of age is around 20%. This figure is 5% in the non-Indigenous community.

While only 0.8% of the non-Indigenous community have follicular trachoma, the Indigenous population rate is 12.5%. 33% of the Indigenous population suffer from cicatricial trachoma. This form is caused by scarring from follicular trachoma. In comparison, only 1% of the non-Indigenous community suffer from this form of the disease.

- 11 Suggest reasons why there are high levels of trachoma in the Australian Indigenous community. What factors have exacerbated this problem?

Extension

- * Research the work Fred and Gabi Hollows did with cataracts.
- * Gabi Hollows is listed as one of Australia's 100 National Living Treasures. What is a 'Living Treasure' and who else is on the list?
- * Our society is ageing. Explain the ramifications of this statement for the extent of eye disease in the community and the impact on society in general.



Gordon Gould

1920–2005

- Gordon Gould was born in New York, USA. He was educated at Union College and Yale University and majored in Physics and Optics.
- During the war, he worked on the Manhattan Project (development of the atomic bomb) and separated uranium isotopes. After the war, he studied optical and microwave spectroscopy.
- In 1957, Gordon created the first written prototype of the laser. However, Theodore Maiman developed the first working laser in 1960 and the patent for lasers was awarded to Bell laboratories. Gordon undertook a 30 year legal battle for the patent rights of the laser and received his first royalties in 1988.
- Lasers are now commonly used in science, industry, communication, medicine and consumer electronics.

Lasers

1 Complete the information using these words: *amplification, atoms, laser, light, reflected, reaction, crystal, energy*.

The word _____ is an acronym for 'light _____ by stimulated emission of radiation'. It is a device that creates a narrow, intense beam of coherent _____.

Atoms or molecules of a _____, gas, liquid or other substance can be excited in a laser cavity to a higher _____ level. Reflective surfaces at both ends of the laser ensure that energy is _____ and increased with each passage. When the material is excited, the _____ discharge in a rapid chain _____ and this produces the intense light beam.

Parts of the laser

2 Label the laser diagram in figure 1 with these words: *mirror, semitransparent mirror, ruby rod, flash lamp*.

Uses of lasers

3 Write an appropriate field or area beside each application in table 1.

Table 1 – Uses of lasers

Field/area	Application of lasers
Medicine	Correction of eye sight
	Bar codes
	Fibre optics
	Fingerprints
	Destroy cancer tumours
	Remote controller for electronic equipment
	Treat heart disease by removing fat from blood vessels
	Measuring sea depth
	Making holographs
	Removal of tattoos, birth marks, hair
	Metals strips cut by laser to stop forgery
	Laser weapons
	Artificial laser-guided stars

Definitions

Laser – Abbreviation for 'light amplification by stimulated emission of radiation'

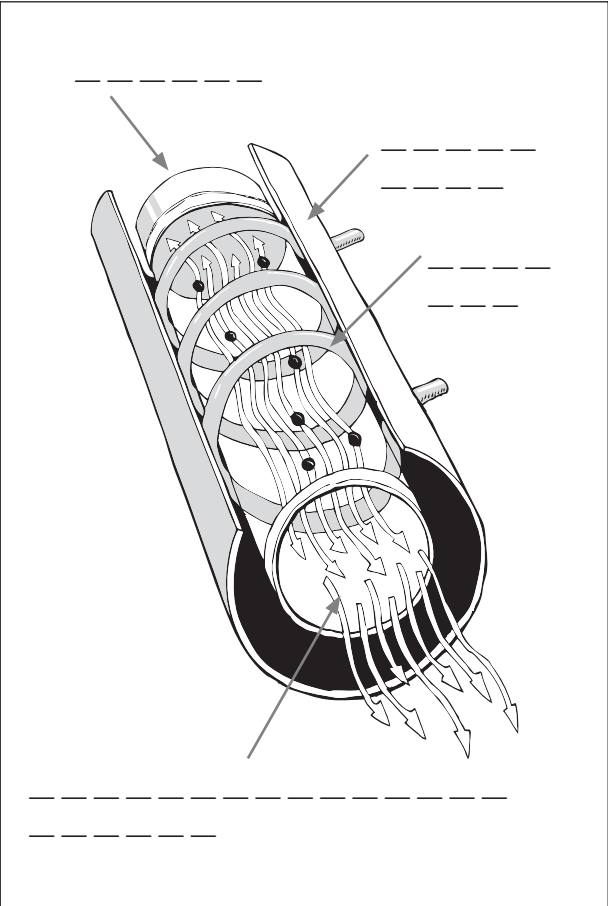
Amplification – Increase in size

CD – Compact disk used to store information

Barcode – Series of bars and spaces used to identify a product

Patent – Legal ownership of an invention

Figure 1 – Laser



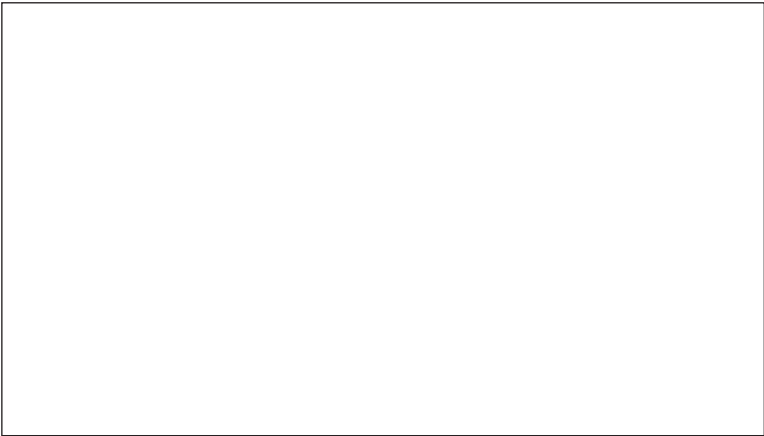
4 Explain what features of lasers make them useful in so many different industries.

DVDs and CDs

A laser can record or read the information on a CD or DVD. Information is cut into the pits of the disk and it is read by a diode laser that examines the reflections from the surface of the disk as it rotates.

- 5 In figure 2, draw a diagram that explains how information is read from a disk. Annotate your diagram with brief explanations. Use these labels: *compact disk, laser source, motor, lens, reflected beam, mirror, focusing lens.*
- 6 Outline the advantages and disadvantages of storing information on DVDs compared to traditional methods.

Figure 2 – Reading information from a disk



Bar codes

Bar codes use a pattern of bars and spaces to identify a product. This pattern is optically scanned and decoded by a computer. A light source is scanned across the barcode, and the intensity of reflected light from the white spaces creates an electronic signal which is decoded by the computer.

- 7 Describe the advantages for retailers of using barcodes instead of individual price tags.
- 8 How can barcodes contribute to the efficient monitoring of stock?
- 9 How do barcodes reduce error, compared to manual data input?
- 10 How can barcodes increase the efficiency of data storage and retrieval?

Figure 3 – Sample barcode



Other uses of lasers

- 11 Outline the applications of lasers in each of the following.
 - a Telecommunication and fibre optics
 - b Brain surgery
 - c Mining and drilling
 - d Laser sights, e.g. rifles and guns
 - e Speed measurements by police
 - f Laser distance meters and surveyors
 - g Missiles and warfare
 - h Earthquake detection
 - i Astronomy

True or false?

- 12 Mark each statement as true or false. Then correct the false statements so that they are true.
 - a High powered lasers can damage your eyes.
 - b Lasers are distorted over long distances.
 - c Lasers cannot clear clogged arteries.
 - d Lasers can be used to store and retrieve huge quantities of information.
 - e Infrared laser guidance made smart bombs possible.
 - f Lasers are not precise enough for eye surgery.
 - g Lasers can cut holes in a diamond.
 - h Lasers produce a thin band of light.

Extension

- * Lasers have become an integral part of modern society. Discuss this statement
- * Research the possible applications of laser technology in the future.
- * Research further how lasers are used in one of the areas listed on this worksheet.



Gustav Nossal

1931–

- Gustav was born in Vienna, Austria. His family immigrated to Australia when he was eight. In 1952, he obtained his medical degree from the University of Sydney and in 1960 a PhD from Melbourne University.
- From 1965–96, Gustav was Director of Eliza Hall Institute of Medical Research.
- While researching immunity, he found that one cell made one antibody and determined the memory region of a cell.
- Gustav worked with the World Health Organisation and produced vaccines for the world’s major diseases.

Disease

Disease refers to any abnormal functioning of the body. Disease can be caused by many factors.

- 1 List three examples of diseases caused by each factor.
 - a Bacteria
 - b Fungi
 - c Virus
 - d Genetic or inherited
 - e Environment
 - f Parasite

Most deadly diseases

Table 1 lists the most deadly diseases in the developing world.

- 2 List the three diseases which have the highest mortality rate.
- 3 Which three diseases have the greatest annual infection rate?
- 4 Explain why these diseases occur predominantly in poorer developing countries.
- 5 Find out how many of the diseases in table 1 exist or cause fatalities in Australia.

Infectious and water related diseases

- 6 Complete the information using these words: *infectious, diseases, million, water, measles, violence, sanitation, 135.*

_____ diseases can be transmitted from one person to another. AIDS, tuberculosis, _____, malaria, diarrhoeal diseases (such as dysentery and cholera) and acute respiratory diseases such as pneumonia are responsible for 90% of all deaths from infectious _____.

Each year, more than 5 _____ people die from water-related diseases. The leading cause of child deaths in the world is diarrhoea. Of the 37 major diseases in developing countries, 21 are related to _____ and sanitation.

The World Health Organisation estimates that 80% of all sickness in the world is attributable to unsafe water and _____. If no action is taken to address the basic need of clean water, as many as _____ million people will die from water-related diseases by 2020. More people die each year from unsafe water than from all forms of _____, including war.

Definitions

Disease – Causes abnormal function of body, e.g. malaria

Vaccine – Dead or diluted portion of a disease given to a person to build antibody levels and resistance

Parasite – Organism that lives off another living being, e.g. flea

Environment – Surroundings, such as climate, soil and water

Table 1 – Most deadly diseases of the developing world

Disease	Annual mortality rate	Annual infection rate
Lower respiratory infections	4 million	–
HIV/AIDS	3 million	39.4 million
Diarrhoea	2.2 million	4 billion
Malaria	1–5 million	300–515 million
Tuberculosis	2 million	8 million
Measles	530 000	30 million
Whooping cough	200 000–300 000	20–40 million
Tetanus	214 000	500 000
Meningitis	174 000	1 million
Syphilis	157 000	12.2 million

Source: World Health Organisation, www.who.int/whr/2004/annex/en/index/htm

Effects of disease

7 List ten effects of disease on our society.

Characteristics of viral diseases

8 Write each viral disease beside the correct characteristics in table 2: *chicken pox, AIDS, influenza, measles, Ross River fever, mumps.*

Fighting disease

Immune sytem of the body

The body has three lines of defence.

9 Referring to figure 1, categorise the following lines of defence as first, second or third:

- * B cells
 - * T cells
 - * tears
 - * mucous
 - * saliva
 - * skin
 - * inflammation
 - * phagocytic white blood cells
 - * antimicrobial substances
- 10 Write a brief paragraph describing the differences between the first, second and third lines of defence.

Controlling disease

Four key factors determine the effectiveness of controlling a disease.

- * Behavioural, e.g. safe sex
- * Modification of environment, e.g. clean water
- * Immunisation, e.g. vaccines
- * Treatment, e.g. antibiotics

11 Referring to each of the four key factors listed above, draw up a table listing control methods for two different diseases.

Vaccines

Vaccines are a suspension of a modified microbe that stimulates the body to produce antibodies to fight the disease.

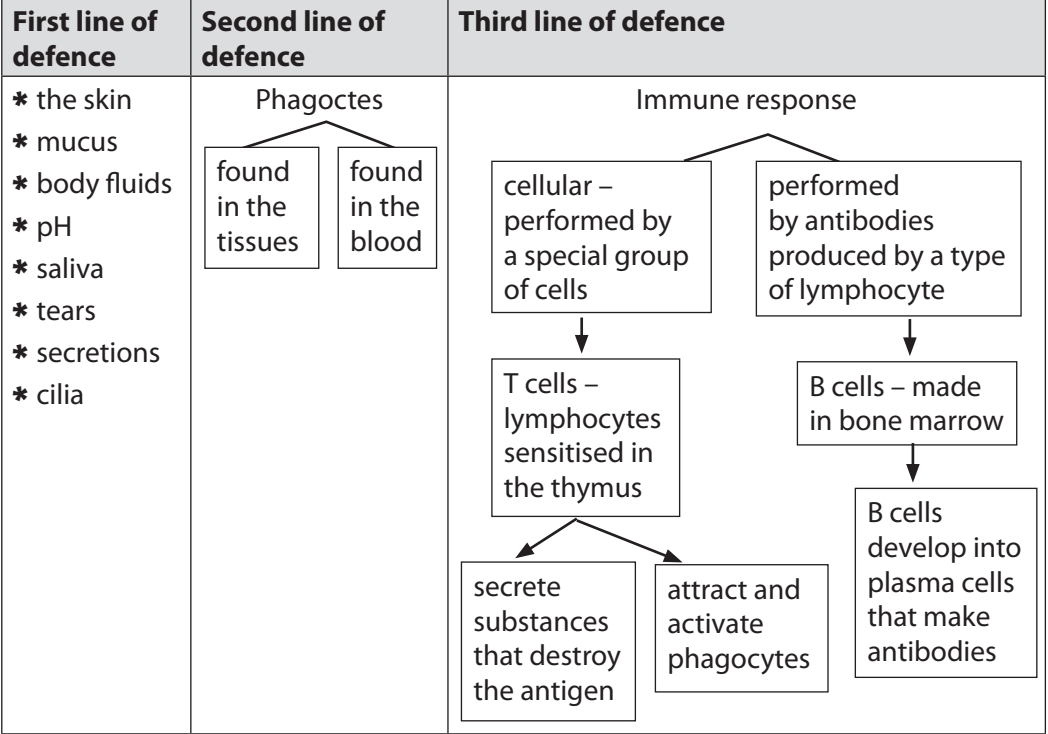
Diseases such as smallpox, rabies, polio, measles, mumps and hepatitis B are controlled with the use of vaccines.

12 List advantages and disadvantages of vaccination.

Table 2 – Characteristics of viral diseases

Disease	Characteristics
	• Fatigue and high temperatures • Spread by blood sucking insects such as the mosquito
	• Skin rash and small red spots • Spread by close contact
	• Aches, pains, high temperatures and runny nose • Spread by water droplets in the air
	• Destroys cells of the immune system • Spread by sexual contact or blood
	• Itchy raised spots on the skin • Spread by close contact
	• Swollen cheeks and neck • Spread by close contact

Figure 1 – Defence systems of the body



Source: Hill J et al, HSC Biology, Pascal Press, Sydney 1990

Extension

- * *Quality of life in developing countries is directly related to the incidence of disease. Discuss this statement.*
- * *Integrated control using education, vaccination, improved water quality and sanitation is the only effective way of controlling disease. Discuss this statement.*
- * *Research the development of vaccines for smallpox, rabies, polio, measles, mumps or hepatitis B.*



Raymond Damadian

1936–

- Raymond Damadian was born in New York, USA. As a child he studied violin at the Juilliard School of Music, then won a scholarship to the University of Wisconsin.
- He obtained his Bachelor of Science degree in 1956 and a Masters degree from Yeshiva University in 1960.
- After serving in the Air Force, he joined the University of New York Medical Centre in 1967.
- Raymond developed the first MRI scanner in 1976 and established the FONAR Corporation for its production.

MRI

The MRI creates 2-D or 3-D images of the body, using magnetic fields and radio waves.

The patient is placed in a strong magnetic field and radio frequency radiation is applied.

Hydrogen atoms in the patient's body resonate and emit radio waves, which a computer interprets into an image.

- 1 Define the term *MRI*.
- 2 Use the information above to label figure 1.

Figure 1 – MRI machine



Definitions

MRI – Magnetic resonance imaging

Radio waves – Form of electromagnetic radiation

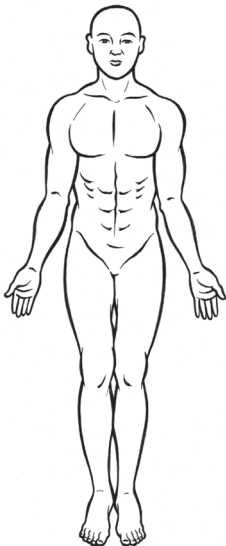
3-D – Images seen in three dimensions

Resonate – To vibrate

Uses of MRI

- 3 Use the list at the right to outline on figure 2 the problems that can be detected by using MRI.

Figure 2 – Problems detected by MRI



Problems that can be identified by MRI

- * Brain tumours
- * Strokes
- * Brain abnormalities of people with dementia
- * Diseases of the pituitary gland
- * Eye or inner ear abnormalities
- * Damage from heart attack or heart disease
- * Blood vessels with blockages
- * Bone and joint damage
- * Reproductive and bladder problems
- * Multiple sclerosis
- * Cancerous tumours in the brain, spinal cord, prostate gland, breast and uterus
- * Infections of the brain, spine and joints

- 4 Explain how an MRI scan differs from an X-ray.
- 5 Why must jewellery and other metal devices be removed before an MRI scan?
- 6 Can people with pacemakers have MRI scans? Explain your answer.

Disadvantages of MRI

MRI scans have no harmful effects because they only use radio and magnetic waves.

- 7 Explain how each factor in table 1 could be a disadvantage when using MRI.

True or false?

- 8 Mark each statement as true or false. Then correct the false statements so that they are true.
- a MRI machines use X-rays.
- b Images can be in two dimensions or three dimensions.
- c MRI machines are not expensive to purchase.
- d The use of MRI machines is rapidly increasing.
- e An MRI machine needs a computer to analyse data.
- f MRI scans are the same as CT scans.
- g An MRI cannot detect infections.

Other uses of magnetic fields

Magnets naturally occur and are commonly used in our society.

- 9 List ten common uses of magnets.
- 10 Write each magnetic device beside the correct use in table 2: *electric motor, compass, junkyard cranes, hard disk, credit cards, television, computer monitors, loudspeakers, microphones.*
- 11 Explain how a compass finds the North Pole.

Table 1 – Disadvantages of MRI

Factor	Why this is a disadvantage
Narrow claustrophobic space where patient has to lie	
Scans take 20–90 minutes	
Patient cannot move while the scan is taking place	
No metals are allowed near the machine	
Devices with radio frequency such as pacemakers cannot be used near the machine	
Scanner makes a hammering noise throughout the procedure	

Table 2 – Other magnetic devices

Device containing a magnet	What it does
	Convert electrical energy into mechanical energy
	Increases sound levels
	Part of a music system where sound is emitted
	Computer screen
	Used in computers to store information
	Lifts metals such as old cars
	Used to watch movies, documentaries and the news etc
	Used to withdraw money or provide a cash advance
	Used to find north

Extension

- * *Magnets play a vital role in modern technology.* Discuss this statement
- * Find out how pigeons use magnetic fields for navigation.
- * Research other applications of electromagnetic waves in medicine.
- * Find out what a superconductor is, and how it is relevant to magnetism.
- * Research the history of the scientific discovery of magnetism.



Evelyn Billings

1918–

- Evelyn was born in Melbourne, Australia. She completed a Bachelor of Medicine and Surgery at the University of Melbourne and a Diploma of Child Health in London.
- In 1953, a Catholic priest asked Evelyn to help his marriage consultancy with family planning issues.
- In 1965, Evelyn, her husband and Dr James Brown developed the ovulation method of contraception. This method can result in a pregnancy rate of less than 1% and more than 100 countries now have trained specialists to advise women on the method.
- Evelyn has nine children of her own.

The Billings method

- 1 Complete the information using these words: *contraception, vaginal, discharges, Billings, intercourse, vulva, percent, harmful, regular.*

_____ discovered a method of natural _____ based on the monitoring of vaginal _____ and changes to the _____.

When using this method, women avoid _____ on all days and for a few days after the appearance of _____ discharges.

The method can result in a pregnancy rate of less than one _____ and does not involve drugs or have any _____ side effects.

Women can use this method during _____ or irregular cycles, breastfeeding or perimenopause.

Definitions

Contraceptive – Method of preventing pregnancy, e.g. the pill, condoms

Billings method – Contraceptive method based on observations of vaginal discharges and changes to the vulva during the oestrous cycle

Menstrual cycle – Approximately 28 day cycle of the female reproductive organs, which prepares the body for pregnancy

STD – Sexually transmitted disease

The menstrual cycle

- 2 Use these words to complete the information below: *lining, three, develop, secretes, oestrogen, ovulation, thickens.*

The menstrual cycle occurs approximately every 28 days and undergoes the following steps:

Step 1: The anterior pituitary gland _____ the follicle-stimulating hormone (FSH), which makes a follicle _____ in the ovary.

Step 2: The ovaries release the hormone _____, which inhibits FSH production.

Step 3: Luteinising hormone is released and causes _____.

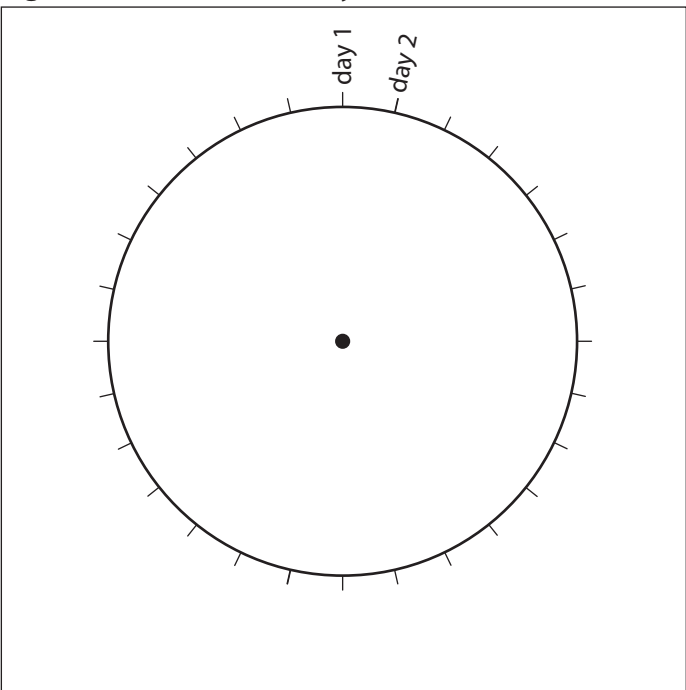
Step 4: The ovary secretes progesterone, which _____ the lining of the uterus and prepares the body for pregnancy. The ovum survives for approximately 12–24 hours and sperm has a life span of approximately _____ days.

Step 5: If pregnancy does not occur, menstruation removes the _____ of the uterus and the cycle continues.

- 3 Complete and label figure 1 to show the events that occur during the menstrual cycle. Use these labels:

- * menstruation
- * ovulation
- * lining of the uterus thickens
- * lining of the uterus is removed
- * pregnancy can occur.

Figure 1 – The menstrual cycle



The Billings method

4 Referring to figure 2, explain how the Billings method works.

Types of contraceptives

Many other methods of contraception are available to men and women. Some are more effective than others at providing protection from sexually transmitted diseases.

- 5 Write each method beside the correct description in table 1: *pill, tubal ligation, condom, diaphragm, natural planning, IUD, hormonal implants and injections, spermicides, morning-after pill, abstinence, vasectomy.*
- 6 Draw up a table in your workbook and list the advantages and disadvantages of each type of contraception listed in table 1.

Sexually transmitted diseases

- 7 Tick the methods in table 1 that provide protection from sexually transmitted diseases.
- 8 List three sexually transmitted diseases, their effects and how they are treated.

The contraceptive pill

The combined contraceptive pill contains two synthetic female hormones, estrogen and progestin. These inhibit or stop ovulation and thicken the cervical mucous to stop the movement of sperm.

The pill is reversible, reduces the incidence of menstrual cramps, heavy bleeding, acne and ovarian cysts and protects against ovarian and endometrial cancer.

- 9 Outline the reasons why the pill is the most popular method of contraception in Australia.
- 10 Explain why the pill should not be taken in conjunction with medications such as antibiotics.

Figure 2 – The Billings method

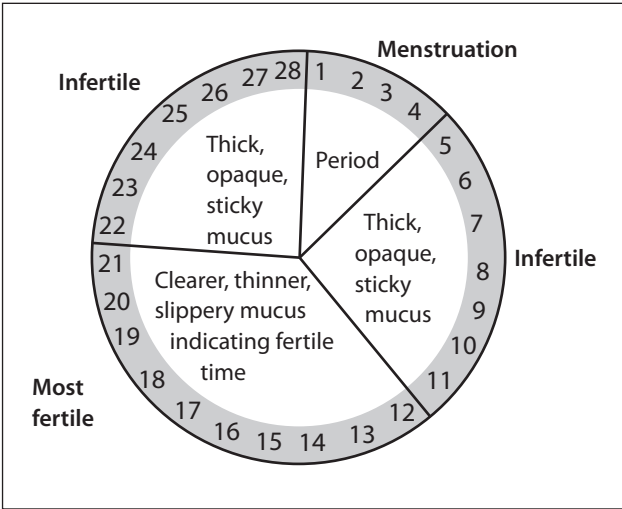


Table 1 – Types of contraceptives

Method	Description
	Rubber cap that fits over the cervix
	Tablets containing hormones that prevent pregnancy
	Hormones implanted or injected under the skin which prevent ovulation
	Substances that are inserted into the vagina before intercourse
	Hormone prescribed by a doctor within 72 hours after sex
	Placed in uterus and prevents or interrupts conception
	Sterilisation of a male
	Latex sheath that fits over the penis
	Monitoring mucous discharges, body temperature changes and rhythm or cycle calculations
	Does not have sex
	Sterilisation of a female

Extension

- * The use of contraceptives remains complicated in some areas. Discuss the impact of tradition, culture, and religious and political groups on the use of contraceptives.
- * Discuss the advantages and disadvantages of the most common types of contraceptives.
- * Research the history of contraceptive devices and techniques.



Robert Oppenheimer

1904–1967

Nuclear fission

Nuclear fission is the process of splitting an atom. This results in the creation of new elements and the release of large quantities of energy. Nuclear fission is used to generate energy in nuclear reactors and atomic bombs.

Uranium is a heavy metal with a high neutron-to-proton ratio. It has two naturally occurring isotopes and disintegrates into lead.

A chain reaction is the splitting of atoms by bombarding them with neutrons. A single neutron can split uranium-235 atoms into barium and krypton, and releases heat and gamma radiation.

- 1 Define the term *nuclear fission* and explain why it releases so much energy.
- 2 Label figure 1 using these words: *uranium-235*, *neutron*, *uranium-236*, *barium-141*, *krypton-92*, *neutrons*.
- 3 Referring to figure 1, explain the process of nuclear fission.

The atomic bomb and Hiroshima

Table 1 lists information about the bombing of Hiroshima in 1945.

- 4 Define *atomic bomb*.
- 5 Describe the size and weight of the bomb.
- 6 What nuclear material was contained in the bomb?
- 7 Why was Hiroshima bombed in 1945?
- 8 Describe the effects of the bombing on Hiroshima and its people.

Table 1 – Effects of atomic bomb used on Hiroshima

Code name	Little Boy
Dropped by	B29 bomber Enola Gay
Size	Length – 3 m; diameter – 0.7 m
Weight	4 tons
Nuclear material	Uranium-235
Energy released	Equivalent to 12.5 kilotons of TNT
Temperature at burst point	Several million °C
Area of total vaporisation	1.3 km
Area of total destruction	2.6 km
Severe blast damage	6.5 km; everything flammable burned; maximum wind velocity 440 m/s
Effects	Thermal heat caused severe burns and loss of eyesight up to 3.5 km from ground zero; a shock wave with wind pressure of 35 tons/m ² blasted the area; fallout rain was laden with radioactive particles and caused radiation poisoning
Death rate	140 000
Number of people on health cards from the incident	352 550

- Julius Robert Oppenheimer was born in New York, USA. He studied physics at Harvard University and undertook his PhD in Germany.
- In 1942, Robert directed the Manhattan Project, which was the development of the atomic bomb. Research cost more than \$2 billion and the first bomb was exploded in New Mexico in 1945.
- Within a month, two atomic bombs were dropped on Japan, who surrendered on 10 August 1945.
- From 1947–52 Robert chaired the US Atomic Energy Commission.
- In 1953, he was accused of communist sympathies, and his security clearance was cancelled.
- He then became Director of the Institute of Advanced Study at Princeton and wrote about the problems of intellectual ethics and morality.

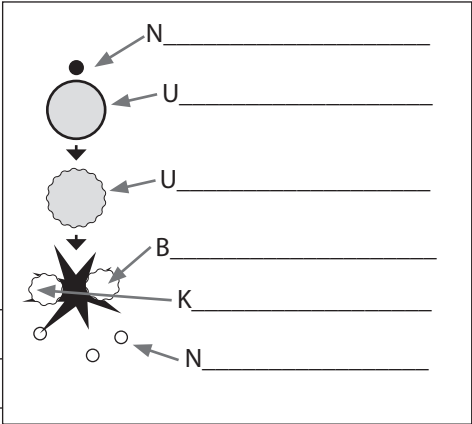
Definitions

Fission – The splitting of an atom

Uranium – Unstable heavy metal used in nuclear reactors

Trinity – First atomic bomb

Figure 1 – Nuclear fission



Effect of radiation exposure

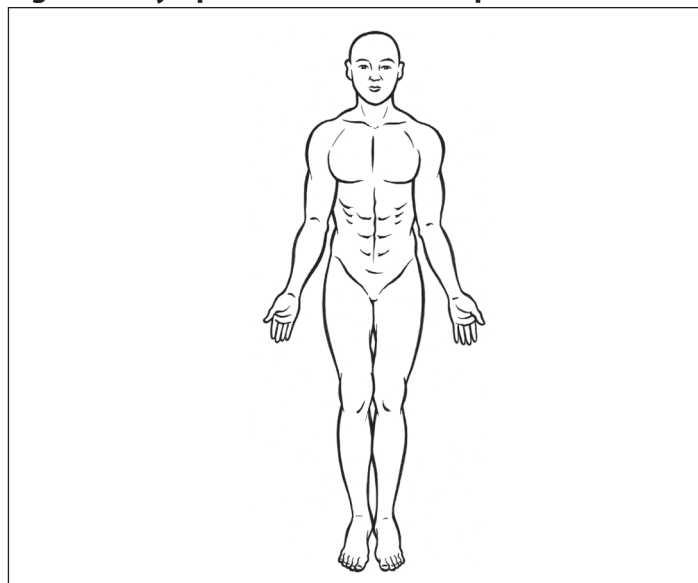
- 9 Use the information below to label figure 2 to show the effects of radiation exposure on a person.

Symptoms of radiation exposure

- | | |
|------------------------------|----------------------------|
| * nausea | * bloody discharge |
| * vomiting | * loss of hair |
| * fatigue | * anaemia |
| * burns | * scarring on burned areas |
| * wounds | * cataracts |
| * headaches | * cancer |
| * low white blood cell count | |

- 10 Radiation can mutate DNA. What could be the long term effects of the bomb at Hiroshima on future generations?

Figure 2 – Symptoms of radiation exposure



Other uses of nuclear energy

Nuclear reactors and energy

More than 30 countries operate nuclear power stations. This provides more than 17% of the world's energy needs. Uranium and plutonium are used in nuclear reactors to produce energy. The heat produced generates steam, which moves turbines and generates electricity. The chain reaction is controlled by boron rods that absorb spare neutrons. Fast breeder reactors use uranium-235 and plutonium-239 and have no moderators. One kilogram of uranium can produce as much energy as 3000 tonnes of coal.

Advantages and disadvantages of nuclear power generation

- 11 Decide whether each factor listed below is an advantage or disadvantage of nuclear power generation. Include reasons for your answers.
- | | |
|---|--|
| a Uranium is a non-renewable resource. | g Disposal of radioactive waste |
| b Reactors are expensive to build. | h Radiation can cause genetic mutations, leukaemia and death. |
| c Less pollution output | i Products can be used to make nuclear weapons. |
| d Reactors have short life spans, e.g. twenty years. | j Uranium is an abundant source of concentrated power. |
| e Fast breeder reactors have no moderators and a high risk of accidents. | k Nuclear power stations are cheap to operate. |
| f Reactors take up less open space than mines. | |

Alternative energy sources

- 12 Describe the advantages and disadvantages of the following alternative energy sources.
- Solar
 - Fossil fuel
 - Wind
 - Tidal
 - Geothermal
 - Biofuel (plant and animal remains)

Extension

- * Discuss the statement: *Nuclear energy should supply our energy needs in the future.*
- * Discuss the impact of nuclear weapons on global society.
- * *Nuclear power plant safety depends on a number of key issues including plant design, operator training and operating procedures.* Discuss this statement in relation to the nuclear disasters at Three Mile Island in America and Chernobyl in the Ukraine.
- * Research the benefits of these applications of radiation: carbon dating, X-rays, cancer treatment, microwaves, geological dating, communication.



Nicolaus Otto

1832–1891

- Nicolaus Otto was born in Holzhausen, Germany. He was trained in business but developed an interest in engine technology for powering horse-drawn carriages.
- In 1864, he formed the company N.A Otto and Cie, which was the first engine company in the world. They invented the first internal combustion engine.
- In 1885, Daimler and Maybach designed and built a motorcycle with an Otto engine.

Importance of the internal combustion engine

Machines such as cars, lawn mowers, chain saws and pumps use internal combustion engines.

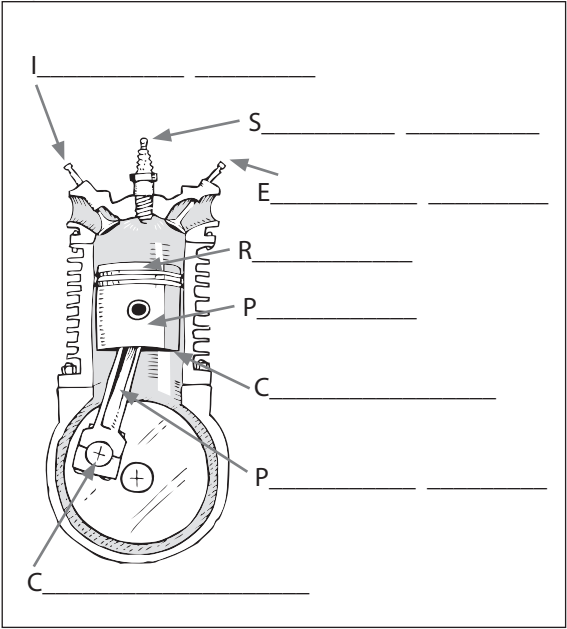
- 1 List five types of machinery that use piston engines.

How a piston works

A spark plug ignites petrol and air in a chamber. This causes an explosion, which moves the piston.

- 2 Use these words to label the parts of the piston in figure 1: *intake valve, rings, cylinder, crankshaft, spark plug, exhaust valve, piston, piston rod*.
- 3 Referring to figure 1, write a brief description of how a piston works.
- 4 Explain the differences between four-cylinder, six-cylinder and eight-cylinder cars.

Figure 1 – Parts of a piston



Definitions

Internal combustion engine

– Machine with pistons that generates power by igniting fuel in chamber

Acid rain – Rain with a pH less than 7 formed from pollutants such as nitrous oxides. Kills plants and corrodes buildings

Greenhouse effect – Warming of the Earth caused by pollutants trapping heat in the atmosphere

Engines and cars

- 5 List 10 benefits provided by cars.
- 6 Cars can create many problems in our society. Outline in table 1 how each factor could be a problem, and then suggest a solution for that problem.

Table 1 – Problems caused by cars

Factor	Why this is a problem	Possible solution
Loss of countryside to roads		
Stress and inactivity of driving		
Isolation of people in outer suburbs		
Road accidents		
Wildlife killed by vehicles		
Heavy freight on roads		
Air and noise pollution		
Depletion of fossil fuels		
Traffic congestion		
High youth accident rate		
Cost of buying and maintaining vehicles		

Cars and air pollution

7 Complete the information using these words: *rain, corrode, extinction, warming, toxic, acid, lead, greenhouse*.

Cars produce many _____ gases, including carbon and sulphur dioxide, nitrous oxide, carbon monoxide and _____. These gases have contributed to smog, _____ rain and the _____ effect.

Rain water reacts with nitrous oxides in the atmosphere to form acid _____. This can destroy forests, lakes and rivers, _____ buildings and cause respiratory problems.

Carbon dioxide accumulates in the atmosphere and prevents heat from leaving the Earth. This can result in global _____ and adverse effects such as climate change, melting of ice caps, increased incidence of diseases and _____ of flora and fauna.

Global warming

8 Referring to the information above, label figure 2 to show how global warming occurs.

Energy consumption

9 Discuss the advantages and disadvantages of each form of transport.

- a Train
- b Car
- c Bicycle
- d Truck
- e Ship
- f Motorcycle

10 Suggest reasons why people prefer to use their own cars, rather than public transport

Improving vehicle use

11 Explain how the following practices could improve our transport systems.

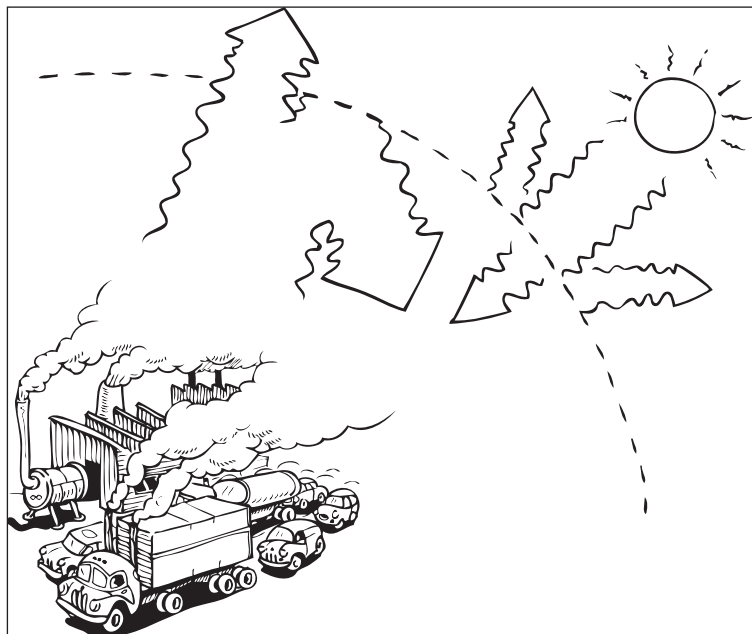
- a Diverting funding to public transport
- b Creating pedestrian-only areas
- c Bike tracks
- d Carpooling
- e Buying a smaller car
- f Driving carefully, not recklessly
- g Engine maintenance
- h Walking
- i Transporting heavy goods by train

True or false?

12 Mark each statement as true or false. Then correct the false statements so that they are true.

- a Fossil fuels are a renewable resource.
- b Global warming will cause climate change.
- c Lead is found in all petrol.
- d Public transport is more energy efficient than using cars.
- e Drinking alcohol affects your capacity to drive.
- f Mobile phone use while driving is a hazard.

Figure 2 – Global warming



Extension

- * *The invention of the car has been one of the greatest environmental problems ever created. List points for and against this argument.*
- * *Youth have one of the highest accident rates of all drivers. Suggest and discuss strategies that could eliminate this problem.*
- * *What strategies could be implemented to reduce our dependency on cars?*



Marie Curie

1867–1934

- Marie Curie was born in Warsaw, Poland. She went to university in Paris where she obtained a degree in Physics and Mathematics and met her husband Pierre Curie.
- In 1898 they discovered the element radium, for which they were awarded the Nobel Prize. Marie found that radium was radioactive and could be used to treat cancer.
- In 1912, Marie she was awarded another Nobel Prize for determining the atomic weight of radium.
- During World War 1, Marie instigated the use of X-ray vans to help injured soldiers. She later played an active role in cancer treatment.
- Marie died of leukaemia in 1934, having been exposed to very high levels of radiation.

Radium

Refer to table 1 to answer these questions.

- 1 What type of element is radium?
- 2 What does the word *radium* mean?
- 3 What is the symbol for radium?
- 4 What element does radium become once it has decayed?
- 5 Why is radium difficult to extract from pitchblende?

Uses of radium

Radium was once used to make hair tonics, toothpaste, ointments and elixirs.

Paint containing radium was used to make the fluorescent dials of compasses, watches and clocks. Many of the women who painted the dials licked their brushes and contracted bone cancer.

Radium has been used to treat cancer and as a source of neutrons for scientific research.

Radium is one million times more radioactive than the same mass of uranium, and decays to form lead.

- 6 Explain how radium can both treat and cause cancer.

Radiation

- 7 Complete the following paragraph using these words: *solid, uranium, radiation, X-ray, unstable, gamma, decay*.

_____ is a form of energy. It is derived from the Sun, devices such as _____ machines and from radioactive elements such as _____.

Radiation travels as waves or energetic particles. It can penetrate water and _____ materials.

Radioactive materials have _____ atoms and release radiation and energy. Half life is the amount of time required for an element to lose 50% of its activity by _____.

The most common forms of radiation are alpha, beta and _____ rays. Gamma particles are the most penetrating, and all three are harmful if inhaled, ingested or absorbed.

Alpha, beta and gamma radiation

- 8 Identify and label the three forms of radiation – alpha, beta and gamma – in figure 1 on the next page.

Table 1 – Characteristics of radium

Discovered	1898
Found in	Pitchblende
Name	Originates from the Latin word <i>radius</i> meaning 'ray'
Symbol	Ra
Classification	Alkaline earth metal
Colour	White luminescent
Atomic number	88
Atomic mass	226.0 amu
Isotopes of radium	Sixteen isotopes are known. Radium-226 is the most common and has a half life of 1620 years
Final product	Disintegrates to form lead
Melting point	700°C
Boiling point	1140°C
Highly radioactive	Emits alpha, beta and gamma radiation
Availability	Small amounts in uranium, e.g. 1 gram of radium per 7 tonnes of pitchblende

Definitions

Radium – Radioactive element used to treat cancer

Luminescent – Emits light

Radioactive – Emits alpha, beta and gamma radiation

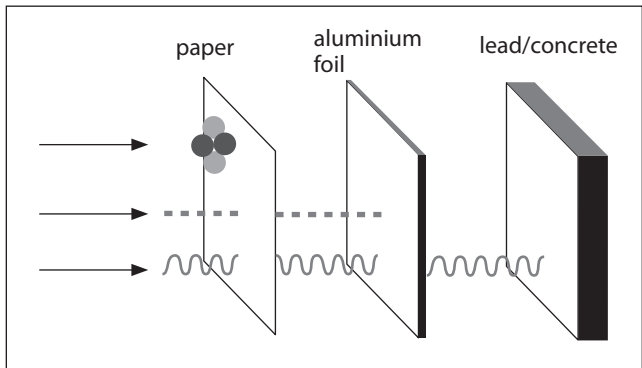
Radiation sickness – Illness caused by exposure to high levels of radiation

Pitchblende – The most common ore of uranium

Radiation and the sun

- 9 These forms of radiation are all emitted by the Sun. Describe the uses of each one.
- | | |
|------------------|----------------------|
| a visible | d infrared |
| b radio | e ultraviolet |
| c X-ray | |
- 10 Explain how the following can be sources of radiation exposure to people.
- | | |
|-------------------------|----------------------|
| a medical X-rays | c cosmic rays |
| b rocks and soil | d radon |

Figure 1 – Alpha, Beta and Gamma radiation



True or false?

- 11 Mark each statement as true or false. Then correct the false statements so that they are true.
- | | |
|--|--|
| a Household appliances such as microwave ovens and television sets produce radiation. | d Rocks and soils can release radiation. |
| b Unstable atoms release radiation. | e Medical and dental X-rays are a form of radiation. |
| c Alpha particles are the most damaging particles. | f 80% of typical human exposure to radiation comes from natural sources such as sunlight. |

Effects of radiation

Acute radiation syndrome or radiation sickness occurs when the body is exposed to high levels of radiation for a short period of time, as in the cases of the bombings of Hiroshima and Nagasaki and the Chernobyl meltdown.

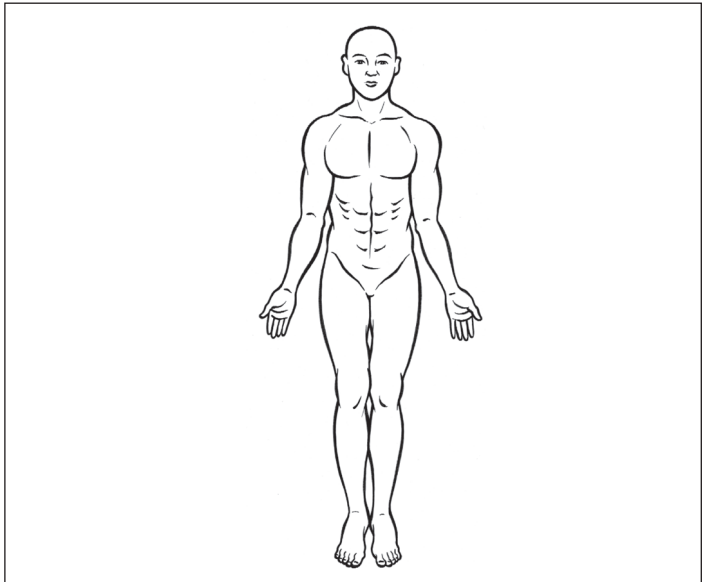
- 12 Referring to the list below, label figure 2 to show the symptoms of radiation sickness.

Symptoms of radiation sickness

- | | |
|-------------------------|-----------------------|
| * Nausea | * Flu-like symptoms |
| * Vomiting | * Infection |
| * Diarrhoea | * Bleeding |
| * Bone marrow depletion | * Cancer |
| * Weight loss | * Cataracts |
| * Loss of appetite | * Decreased fertility |
| | * Death |

- 13 Suggest why the damage from exposure to radiation may not be apparent for many years.
- 14 Why are children and embryos more sensitive to radiation than adults?

Figure 2 – Symptoms of radiation sickness



Benefits of radiation

- 15 Discuss the benefits of radiation in each field or area.
- | | |
|--|--|
| * Hospitals | * Nuclear power plants |
| * Research laboratories | * Smoke detectors |
| * Industry and construction sites | * Cooking facilities, e.g. microwave ovens |
| * Diagnosing and treating illnesses such as cancer | * Carbon-14 dating |
| * Sterilising equipment | * Plants and photosynthesis |
| * Irradiating food | |

Extension

- * Discuss the relevance of the Chernobyl nuclear reactor meltdown to the future use of nuclear energy.
- * *Radiation is essential for life.* Discuss this statement.
- * Outline the benefits and problems resulting from the use of radioactive materials.



Barry Marshall

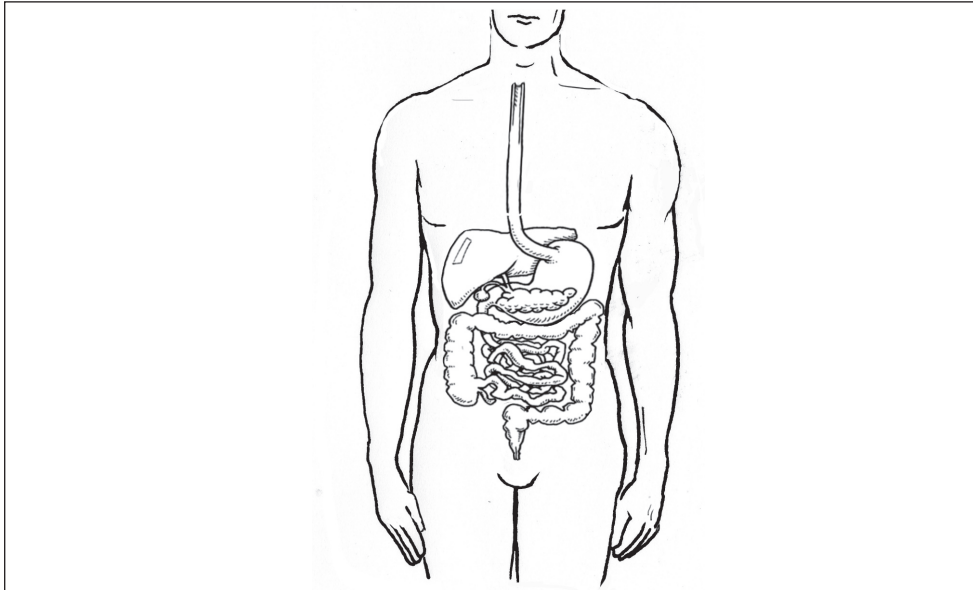
1951–

- Barry Marshall was born in Kalgoorlie, Australia. In 1974, he completed his medical degree at the University of Western Australia, then worked with Robin Warren at the Royal Perth Hospital.
- Together, they showed that *Helicobacter pylori* caused 90% of duodenal ulcers and 80% of gastric ulcers and was a contributing factor in stomach cancer.
- In 2005, they were awarded the Nobel Prize for their discovery.

Digestion

The digestive system breaks down food to supply us with our nutritional requirements.

Figure 1 – The digestive system



Definitions

Digestion – Physical, chemical and microbial breakdown of food into a form that the body can use.

Disease – Factors that cause the abnormal functioning of the body, e.g. malaria

Helicobacter pylori – Bacterium that causes stomach ulcers

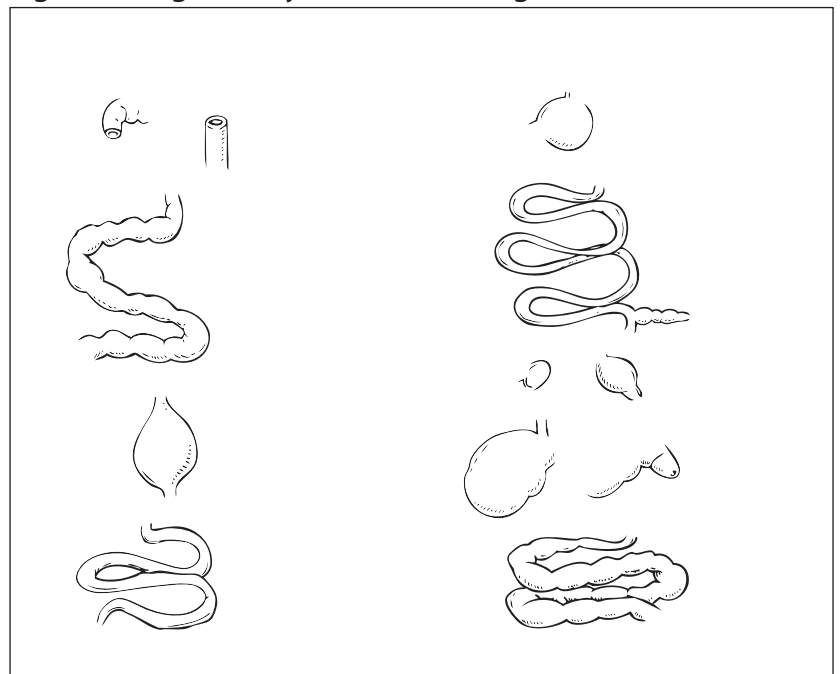
- 1 Label the parts of the digestive system in figure 1 using these words: *pancreas, stomach, mouth, appendix, oesophagus, liver, large intestine, small intestine, anus, gall bladder.*

Functions of the digestive system

- 2 Describe the function of each part of the digestive system listed below.

- | | |
|--------------------------|--------------------------|
| a mouth | e pancreas |
| b oesophagus | f liver |
| c stomach | g large intestine |
| d small intestine | h anus |

Figure 2 – Digestive systems of a monogastric and ruminant



Digestive systems of monogastrics and ruminants

Monogastrics such as humans and chickens have one stomach. They use physical and chemical enzymes to break down food.

Ruminants such as sheep and cattle have four stomachs. They use physical, chemical and microbial action to break down food.

- 3 Cut out figure 2 and place the components of each digestive system in the correct order.
- 4 Outline the similarities and differences between the two digestive systems.
- 5 Name three other ruminant and monogastric animals.

Diseases of the digestive system

Ulcers

- 6 Complete the information using these words: *pain, ulcer, stomach, surgery, three, chemicals*.

An _____ is a sore or open painful wound. Ulcers cause _____, may require major _____ and can cause life-threatening bleeding and peritonitis.

Peptic ulcers occur in the _____ or the duodenum, and are caused by a bacterium.

One in _____ people over the age of 40 is infected with the bacterium. It lives in the lining of the stomach and releases _____ that cause irritation and inflammation.

Symptoms of stomach ulcers

- 7 Referring to the list below, label figure 3 to show the symptoms of stomach ulcers.

Symptoms of stomach ulcers

- | | |
|---|--------------------------|
| * Stomach pain with sharp aches between the breast bone and navel | * Hiccups |
| * Loss of appetite | * Weight loss |
| * Nausea | * Vomiting with blood |
| * Burping | * Bloody bowel movements |

Diagnosis and treatment of stomach ulcers

- 8 Write each method of diagnosis beside the correct meaning: *C-14 breath test, X-rays, blood test, endoscopy, biopsy*.

a Inserting a tube with a small camera into the digestive system

b A sample of blood is analysed for disease.

c Tissue sample is taken during an endoscopy and tested in the laboratory

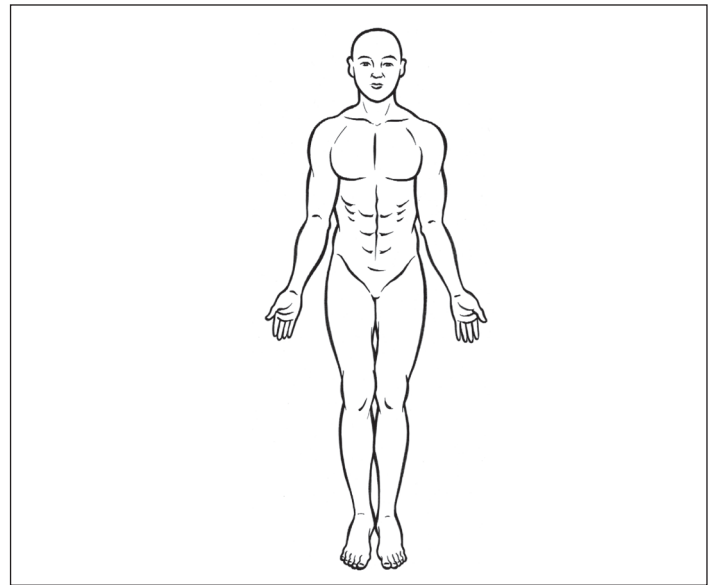
d A small amount of radioactive carbon-14 is swallowed, and the air exhaled from the lungs is tested. If bacterium is present, it converts urea into carbon dioxide.

e Radiation used to view the digestive system

Once the bacterium has been identified, antibiotics can be prescribed to treat the problem.

- 9 For almost 100 years, doctors believed that stress, spicy foods and alcohol caused most ulcers. How do you think the discovery of the effects of the *Helicobacter pylori* bacterium impacted on the recovery rate of victims?
- 10 Explain what antibiotics are and why they would be more effective than traditional methods of treating stomach ulcers.

Figure 3 – Symptoms of stomach ulcers



True or false?

- 11 Mark each statement as true or false. Then correct the false statements so that they are true.
- a** Appendicitis can be fatal.
- b** Methane is produced by ruminants and is a major cause of global warming.
- c** Fibre is not essential in our diet.
- d** Ruminants have four stomachs and use microbes to break down food.
- e** All diseases of the digestive system can be cured.
- f** Diarrhoea is one of the major causes of death in developing countries.

Extension

- * Discuss the causes and control of five major diseases of the digestive system.
- * Compare the structure and function of the digestive system of two native species of carnivore and herbivore.



William Farrer

1845–1906

- William Farrer was born in Westmoreland, England in 1845. He studied medicine then emigrated to Australia in 1870 because he had tuberculosis and needed to live in a dry climate.
- From 1875–86 he worked with the Department of Lands as a surveyor.
- In 1898, William joined the Department of Agriculture and selectively bred wheat for quality, yield, ease of harvesting and disease resistance.
- He developed Federation wheat, a good quality, high yielding variety with short stems and early maturity. From 1910–25 it was the most popular wheat grown in Australia, and increased wheat production four-fold.

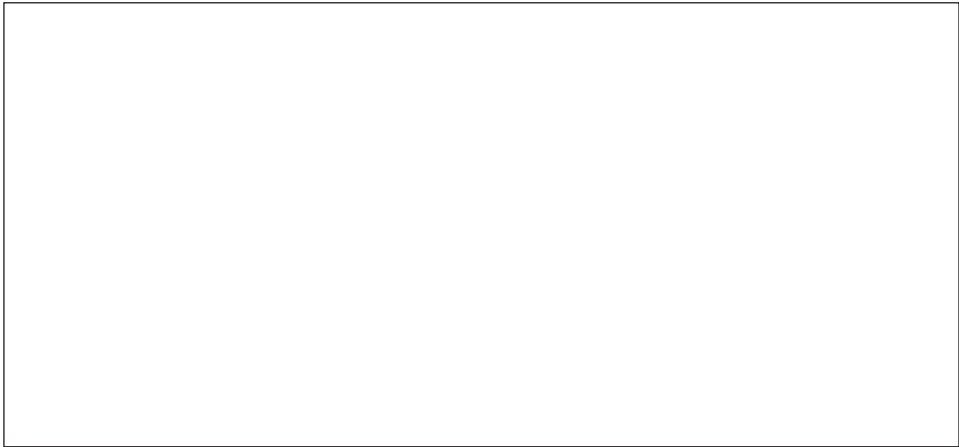
Importance of wheat

- 1 List four products derived from wheat.

Structure of a wheat plant

- 2 In figure 1, draw and annotate a diagram of the parts of a wheat plant and wheat grain. Use these labels: *tillers, leaves, spike, flag leaf, fibrous root system, spikelet, floret, wheat grain, brush, pericarp, aleurone, embryo, endosperm.*

Figure 1 – Structure of a wheat plant



Definitions

Wheat – Annual grass used to make flour
Rust – Fungal disease of wheat
Disease resistance – A plant which is not affected by a disease

- 3 Explain why dwarf wheat is superior to long-stemmed varieties for ease of harvesting.
- 4 Outline how breeding long, thin stems can increase drought resistance.

Calendar of operations

- 5 Referring to figure 2, describe the main operations that occur on a wheat property.

Figure 2 – Calendar of operations for growing wheat

Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
prepare ground				sowing			spray weeds			harvest	

Wheat yields over time

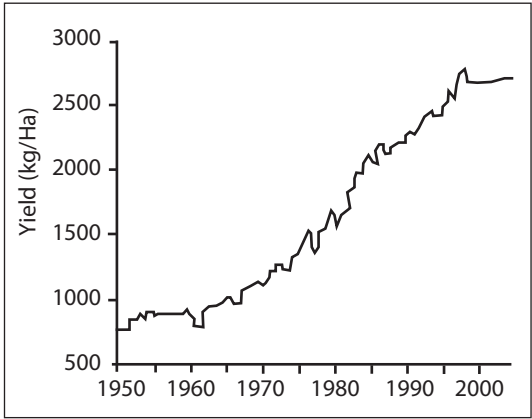
Graph 1 shows wheat yields in developing countries 1950–2004.

- 6 Discuss the shape of the graph and explain which social and technological developments have produced this result.

Factors affecting wheat yields

- 7 Outline how each factor affects wheat yields.
- a Climate
 - b Soil type
 - c Weeds
 - d Pests
 - e Diseases
 - f Variety used
 - g Management practices, e.g. fertiliser and sowing rate

Graph 1 – Wheat yields in developing countries, 1950–2004



Source: Food and Agriculture Organisation of the United Nations FAO Statistical yearbook 2005 and FAO STAT

Diseases of wheat

Stem rust

Stem rust occurs worldwide and is caused by the fungus *Puccinia graminis*. Rust requires two separate plants to complete its lifecycle. Wheat and barberry (*Berberis vulgaris*) are common hosts. Rust can significantly reduce quality and yields. It is controlled by using early maturing or rust-resistant varieties of wheat.

- 8 What climatic factors do you think promote fungal growth?
- 9 How does the removal of secondary hosts increase wheat yields?

Preventing disease

10 Outline in table 1 how each procedure could reduce the incidence of disease.

Table 1 – Procedures to reduce the incidence of disease

Procedure	Effect
Using resistant varieties	
Good hygiene, e.g. removing residues	
Removing weeds or alternate hosts of disease	
Using certified disease free seed	
Crop rotations to break disease and pest cycles	
Timing of operations, e.g. warm, humid conditions favour stem rust	
Management practices, e.g. use of fungicides	
Integrated pest management – using many methods to control a problem	

True or false?

- 11 Mark each statement as true or false. Then correct the false statements so that they are true.
 - a Rust is caused by a bacterium.
 - b Protein content and hardness determine the end use of wheat.
 - c Short stemmed wheat is easier to harvest.
 - d Wheat is not a major crop in the world.

Technology and wheat production

- 12 Outline the effect of the following technology on wheat production.
 - a Genetic engineering
 - b Herbicides
 - c Pesticides
 - d Fertilisers
 - e Machinery, e.g. harvesting machines
 - f Transportation, e.g. rail freight
 - g Global positioning systems
 - h Milling factories and production lines

Extension

- * Discuss the importance of wheat and its products to Australian society, both past and present.
- * Research two other Australian agricultural products and the issues affecting their production.



Crossword of legendary science

Across

- 1 Helicobacter pylori is a bacterium that causes ____
- 3 Method of preventing pregnancy
- 5 Machine with pistons that generates power by igniting fuel in a chamber: internal ____ engine
- 8 Radioactive element used to treat cancer
- 9 Deformity caused by the incomplete closure of the spinal column
- 11 Electric and magnetic fields that travel at the speed of light: electromagnetic ____
- 12 Brain ____ can be identified by an MRI machine
- 13 Thin, glass tube through which light travels
- 16 Bacterial disease of the eye that can result in blindness
- 17 Developed the atomic bomb
- 20 Chemical made up of repeating units
- 21 Invented the laser
- 23 Researched immunity and antibody production and developed vaccines for diseases
- 25 Invented nylon
- 26 Invented the first internal combustion engine
- 30 Developed the light principles of fibre optics
- 31 Invented the artificial hearts Jarvik 7 and Jarvik 2000
- 34 Developed a method of birth control that is used in more than 100 countries
- 35 The Earth science that studies fossil organisms and related remains
- 36 Scientist who developed spray-on skin for burn victims
- 37 Mechanical device with many functions, including storing and analysing data
- 38 Geologist and palaeontologist who researched changes in Australian plants and animals
- 39 A common antibiotic

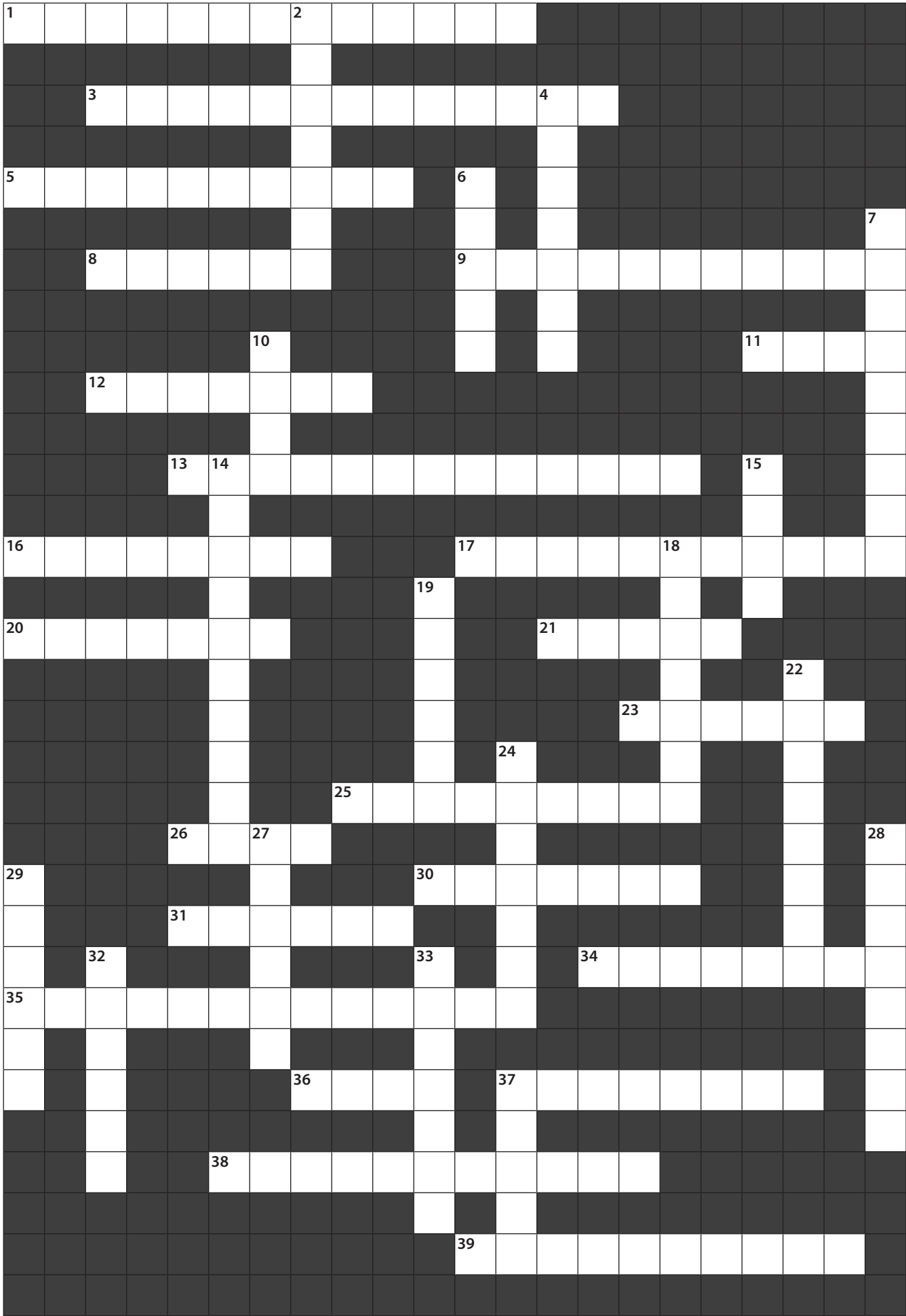
Down

- 2 Unstable heavy metal used in nuclear reactors
- 4 Dead or diluted portion of a disease, given to a person to build antibody levels and resistance
- 6 Abbreviation for 'light amplification by stimulated emission of radiation'
- 7 Small electrical device used to stimulate the heart to beat
- 10 Fungal disease of wheat
- 14 Fuel used in a rocket
- 15 The human body's largest organ
- 18 Married couple who restored the sight of people, worldwide
- 19 Developed good quality, high yielding, drought resistant wheat
- 22 Developed the Magnetic Resonance Imaging scanner
- 24 Discovered that folic acid prevents the birth defect spina bifida
- 27 The founder of computers and computer science
- 28 Discovered the bacterium that causes stomach ulcers
- 29 Inventor of the mobile phone
- 32 Received the Nobel Prize for the isolation and mass production of penicillin
- 33 Pioneer in rocket construction and space flight
- 37 Discovered the element Radium

Word list

BILLINGS	FARRER	NOSSAL	PROPELLANT	TUMOURS
CAROTHERS	FLOREY	OPPENHEIMER	RADIUM	TURING
COMBUSTION	GODDARD	OPTICAL FIBRES	RUST	TYNDALL
COMPUTER	GOULD	OTTO	SKIN	URANIUM
CONTRACEPTIVE	HOLLOWS	PACEMAKER	SPINA BIFIDA	VACCINE
COOPER	JARVIK	PALAEONTOLOGY	STANLEY	VICKERS-RICH
CURIE	LASER	PENICILLIN	STOMACH ULCERS	WAVE
DAMADIAN	MARSHALL	POLYMER	TRACHOMA	WOOD

Crossword of legendary science *cont'd*





Selected answers

Robert Goddard (pages 4–5)

- 1 Chinese, gunpowder oxygen, rocket, London, range, space, astronaut, Cold War, missiles, man
- 2 Telecommunications, research, global positioning systems, weather forecasting
- 3 Power and prestige
- 4 Can attack another country from far away, therefore a greater threat
- 5a 33 360 000 Newtons b 80 800 kg c Needed to push through atmosphere and against gravity
- 7a 27 359 km per hour b liquid hydrogen d 14 crew

Patricia Vickers-Rich (pages 6–7)

- 1 Microbes and invertebrates
- 2a 450 mya b 225 mya c 225 mya d 600 mya e 600 mya f 400 mya
- 3 Greater number and variety have developed with time
- 4 (In vertical order) 2, 5, 3, 1, 4
- 6 South America, Africa, India, Antarctica and Australia
- 7 Antarctica
- 8 Moved closer to the equator
- 9 *Mesosaurus*, *Cynognathus*, the fern *Glossopteris* and *Lystrosaurus*
- 10 Similar structural features
- 12 Up to one third of Australia was once covered by sea

Alan Turing (pages 8–9)

- 1 Babbage, person, mechanical, engine, room, hundred, integrated, smaller, expensive, use
- 4 United States of America
- 6 Central processing unit, input device (e.g. keyboard and mouse), memory in the computer RAM or ROM and long term memory on a disk
- 7 (In table order) CD-ROM, mouse, modem, virus, RAM, ROM, hard drive, motherboard, CPU, monitor, DVD+RW, Internet
- 10 a 12–14 years b 5–8 years c 12–14 years
- 13 Flexible pathways for information, network extends beyond the boundaries of any one nation or organisation

Fiona Wood (pages 10–11)

- 1 Shape, appearance, protection, exchange of water and minerals and keeping cool
- 3 Burns, skin, light, peeling, scalds, scarring, electricity, chemicals, muscle, movement
- 4 First degree burns cause the least damage and third degree burns cause the greatest damage
- 5 More easily damaged
- 8 Does not help the burn and can increase pain level
- 10 Greater exposure to the sun and many locations have beaches where people sunbake. Also many of these regions have populations with fair skin
- 12 People do not protect their skin and spend more time in the sun
- 13 Hat, sunblock cream and protective clothing

Robert Jarvik (pages 12–13)

- 2 disease, million, leading, coronary, attack, arteries, oxygen, strokes, burden
- 5 Insufficient donors
- 7a angiography b medication c bypass surgery d pacemaker e blood pressure f defibrillator g heart transplant h electrocardiogram

Martin Cooper (pages 14–15)

- 1 mobile, electronic network, radio, two, rate, contracts, communication, tasks
- 2 1.2 kg
- 3 35 minutes
- 4 10 hours
- 8 Exponential growth
- 9 Inexpensive, quick, simple and efficient and perform a multitude of tasks
- 12 Replaced with new improved versions and relatively inexpensive
- 13 Toxic wastes in mobile phones can contaminate environment

Fiona Stanley (pages 16–17)

- 1 Spina, birth, nervous, environmental, family, follic, conception, ultrasounds, incidence
- 2 Birth defect with incomplete formation of the brain, spinal cord and nervous tissue
- 3 Genetic and environmental factors
- 4 In the first three months
- 6 Carbon, hydrogen, nitrogen and oxygen
- 7 Insufficient grains, meat, fruit and vegetables
- 10 Reduced incidence of disease
- 12 Anorexia – starvation; Bulimia – inducing vomiting to lose weight. In developed countries, thinness is perceived as beautiful and many females try to lose as much weight as possible

John Tyndall (pages 18–19)

- 1 fibres, light, millions, metre, telecommunications, laser
- 7 Data transmission
- 8 Small size, flexible and lightweight; Cost
- 10 So that hackers cannot interfere with their computer systems
- 11a True b False c False d True e False f False

Wallace Carothers (pages 20–21)

- 1 nylon, coal, polymer, silk, stockings, parachutes, 1941, production, demand
- 2 Carbon, oxygen, hydrogen, nitrogen
- 3 Consists of many repeating units
- 7 Sweat is not absorbed and sticks to the person, which makes them hot and uncomfortable
- 8 Nylon is highly flammable and melts onto the skin of the fire victim
- 10 Tensile strength declines with age
- 11 Must be regularly replaced or safety equipment may not work when needed
- 12 a True b True c False d True e True f True g False

Sir Howard Florey (pages 22–23)

- 1 Antibiotics, fungus, pneumonia, viruses, wall, allergic
- 2 Carbon, hydrogen, oxygen, nitrogen and sulphur
- 3 High temperatures can kill the fungus
- 4 a True b False c True d True e True f True
- 6 Resistance levels are rapidly increasing
- 7 Susceptible microbes are killed and resistant species produce more resistant offspring
- 8 Over-use increases amount of resistance in population
- 9 Microbe may survive, and the problem continues
- 10 a Kenya 25% b Hong Kong 30% c Mexico nearly 50% d Hungary 55%
- 11 Depends on the amount of antibiotics prescribed and if they are used correctly
- 13 No methods of control
- 15 a Prevents disease entering a region b Increases antibodies in the body, ready to protect the person from disease c Encourages good hygiene and sanitation, which helps prevent the spread of disease d Helps control disease e Helps keep people clean and reduce the spread of disease

Fred and Gabi Hollows (pages 24–25)

- 1 eye, light, tissue, cornea, iris, pupil, cones, rods, optic nerve
- 3 blindness, 2700 AD, 400 million, Africa, Asia, America, Islands, Chlamydia trachomatis, adulthood, pain, eyelids, flies, gnats, antibiotics
- 5 Scarring of the upper eyelids, rubbing of the cornea and ulcers affect the person's ability to see
- 7 (In table order) surgery, hygiene, sanitation, education, adequate diet, medicine
- 8 Highest levels occur in poorer countries such as in Africa; often countries with hot, dry climates that have minimal hygiene and sanitation. Cost of control, lack of clean water and sanitation
- 11 Lack of hygiene and sanitation, lack of education and poor living conditions
- 12 Lack of hygiene and sanitation, lack of education and resources

Gordon Gould (pages 26–27)

- 1 laser, amplification, light, crystal, energy, reflected, atoms, reaction
- 4 flexible, versatile, concentrated energy source, can transmit huge amount of information over long distances without distortion
- 7 Less time, labour and money
- 8 Computer can automatically order more stock when numbers decline
- 9 Less human error
- 10 Automatic
- 12 a True b False c False d True e True f False g True h True

Gustav Nossal (pages 28–29)

- 2 Lower respiratory infections, HIV/AIDS, diarrhoea
- 3 Diarrhoea, malaria, whooping cough
- 4 Poor hygiene and sanitation
- 6 Infectious, measles, diseases, million, water, sanitation, 135, violence
- 8 (In table order) Ross River fever, measles, influenza, AIDS, chicken pox, mumps

Raymond Damadian (pages 30–31)

- 1 Magnetic resonance imaging is a machine that creates 2-D or 3-D images of the body using magnetic fields and radio waves
- 4 Safer, as it uses radio waves rather than X-rays
- 5 Magnetic field will pull at the metal devices
- 6 No, a pacemaker contains metal which is affected by the magnetic field
- 8 a False b True c False d True e True f False g False
- 11 A compass is magnetic and its south end is attracted to the North Pole

Evelyn Billings (pages 32–33)

- 1 Billings, contraception, discharges, vulva, intercourse, vaginal, percent, harmful, regular
- 2 secretes, develop, oestrogen, ovulation, thickens, three, lining
- 4 Intercourse should not take place after ovulation
- 5 diaphragm, pill, hormonal implants and injections, spermicides, morning-after pill, IUD, vasectomy, condom, natural planning, abstinence, tubal ligation
- 9 Quick, simple, inexpensive and effective
- 10 Will not prevent conception when taken at the same time as antibiotics

Robert Oppenheimer (pages 34–35)

- 1 Breakdown of an atom with the release of energy from within its bonds
- 4 Explosive device which produces large quantities of energy from the breakdown of an atom
- 5 3 metres in length, 0.7 metres in diameter, weighing 4 tons
- 7 To make Japan cease fire and stop the war
- 8 Mass death and destruction
- 10 Deformities that will appear in future generations

Nicolaus Otto (pages 36–37)

- 1 e.g. car, motorbike, steam engine, ship engine
- 4 The number of cylinders refers to how many pistons are in the engine. The more cylinders, the more power the engine has.
- 5 Quick, simple, reliable transport
- 7 toxic, lead, acid, greenhouse, rain, corrode, warming, extinction
- 10 Quick, simple and convenient
- 12 **a** False **b** True **c** False **d** True **e** True **f** True

Marie Curie (pages 38–39)

- 1 Radioactive element
- 2 From Latin *radius* meaning 'ray'
- 3 Ra
- 4 Lead
- 5 Only 1 gram found in 7 tons of pitchblende
- 6 Kills cancer cells
- 7 radiation, X-ray, uranium, solid, unstable, decay, gamma
- 11 **a** True **b** True **c** False **d** True **e** True **f** True
- 13 May affect chromosomes and only become apparent in old age or in the next generation
- 14 Cells are still actively dividing

Barry Marshall (pages 40–41)

- 4 Monogastrics have one stomach and ruminants have four; monogastrics use enzymes to break down food and ruminants use microbes and enzymes to break down food
- 6 Ulcer, pain, surgery, stomach, three, chemicals
- 8 **a** endoscopy **b** blood test **c** biopsy **d** C-14 test **e** X-rays
- 9 More people can be helped and cured
- 10 Specific chemicals that will kill the microbe infection
- 11 **a** True **b** True **c** False **d** True **e** False **f** True

William Farrer (pages 42–43)

- 1 Grain, flour, hay, livestock feed
- 3 Does not fall over easily
- 4 Reduces surface area, transpiration and water use
- 8 Wet, humid conditions
- 9 Without the secondary host, the disease will die
- 11 **a** False **b** True **c** True **d** False

Crossword of legendary science (pages 44–45)

Across

- 1 stomach ulcers; 3 contraceptive; 5 combustion; 8 radium;
- 9 spina bifida; 11 wave; 12 tumours; 13 optical fibres;
- 16 trachoma; 17 Oppenheimer; 20 polymer; 21 Gould;
- 23 Nossal; 25 Carothers; 26 Otto; 30 Tyndall; 31 Jarvik; 34 Billings;
- 35 palaeontology; 36 Wood; 37 computer; 38 Vickers-Rich;
- 39 penicillin

Down

- 2 uranium; 4 vaccine; 6 laser; 7 pacemaker; 10 rust; 14 propellant;
- 15 skin; 18 Hollows; 19 Farrer; 22 Damadian; 24 Stanley;
- 27 Turing; 28 Marshall; 29 Cooper; 32 Florey; 33 Goddard;
- 37 Curie

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Fiona Wood – Spray-on skin for burns	Evelyn Billings – Contraceptives
Robert Jarvik – Artificial heart	Robert Oppenheimer – Atomic bomb
Martin Cooper – Mobile phone	Nikolaus Otto – Four-stroke internal combustion engine
Fiona Stanley – Pregnancy, folic acid and spina bifida	Marie Curie – Radium
John Tyndall – Fibre optics	Barry Marshall – Digestive system, ulcers and microbes
Wallace Carothers – Nylon	William Farrer – Wheat and genetics
Howard Florey – Penicillin	
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ABN 50 074 266 023
108 Main Rd
Clayton South
VIC 3169
+61 3 9558 4433
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